



UNIVERSITÄT ZU LÜBECK
INSTITUT FÜR ROBOTIK
UND KOGNITIVE SYSTEME

Joint Locking Mechanisms for Soft Robotic Finger

Master's Thesis

From Guru Nishaanth Saravanan

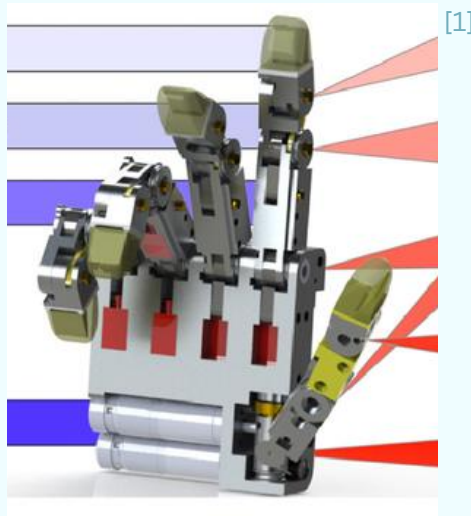
1st supervisor: Dr. Julia Starke

2nd supervisor: Dr. Georg Schildbach

Introduction

Rigid Robotics

- High precision and load capacity
- Poor compliance
- Challenge in interaction with fragile objects and unstructured environment



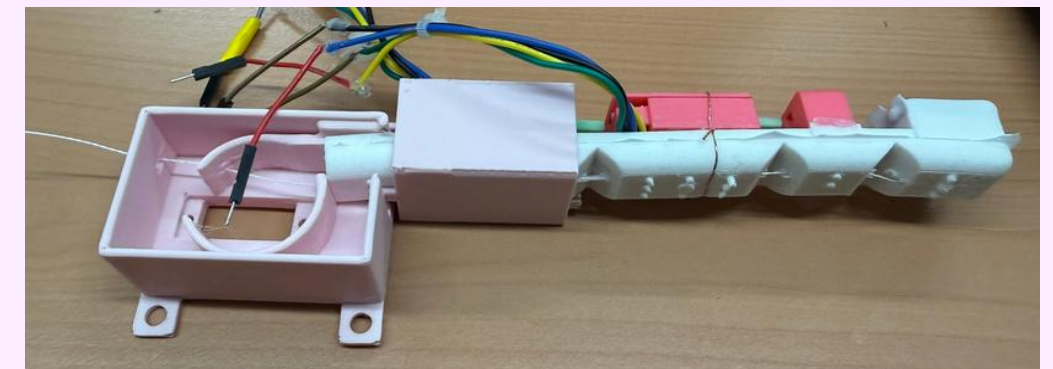
Soft Robotics

- High compliance
- Safer robot-human interaction
- Adaptive grasping with fragile and irregular objects.
- Low precision and load capacity



Hybrid - This Thesis

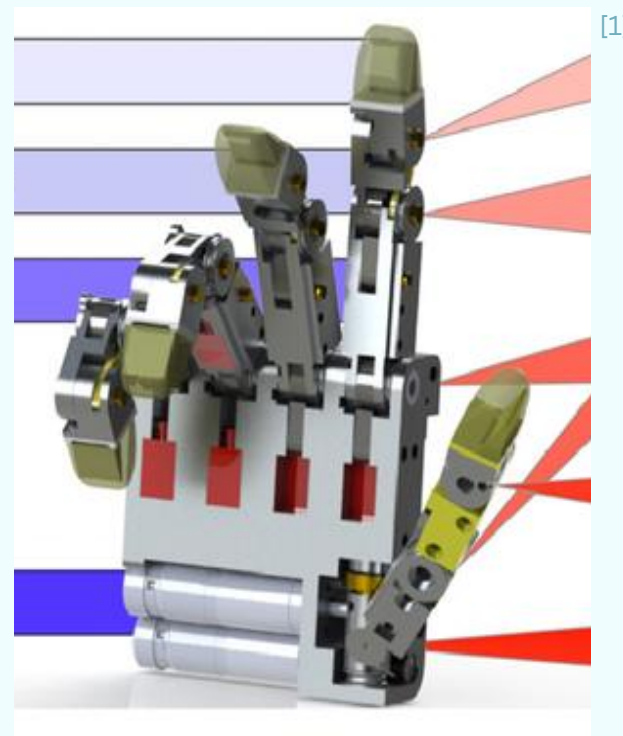
- Soft Body + locking mechanism
- High compliance
- Safer robot-human interaction
- Adaptive grasping with fragile and irregular objects.
- Higher precision
- Better load capacity



Introduction

Underactuated Robotics

Drive multiple joints from a single actuator through passive tendon or linkage coupling.



[1]

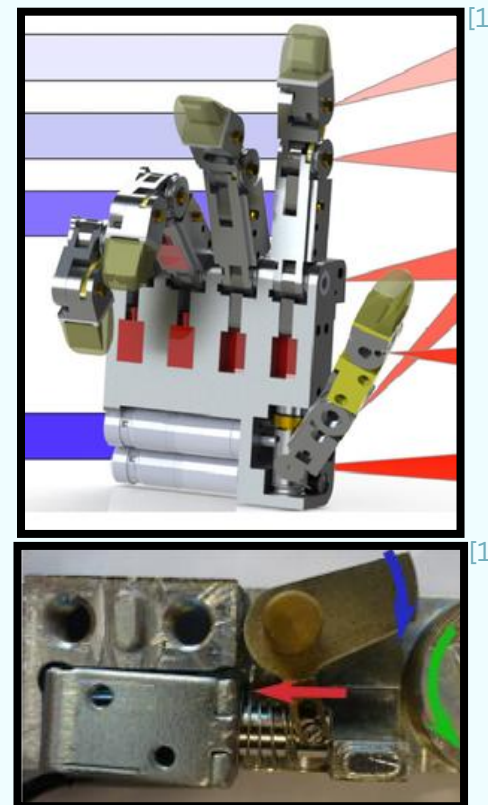
Need for Locking mechanisms

- Underactuated robotic had uses locking mechanisms to compensate the lack of actuators.
- These mechanisms locks the fingers on certain positions to have more dexterity by having multiple configurations

Literature Review

UT Hand 1

- This hand is a rigid, underactuated hand that uses a motor to control each finger's flexion.
- It has an embedded solenoid-based pin-locking mechanism and is used to stop the joint rotation; therefore, locking the joint is in a single position.



Mag-Net finger

- This soft finger is cast using the silicone casting method and is capable of performing flexion motion.
- It has embedded electromagnetic coils



The gap

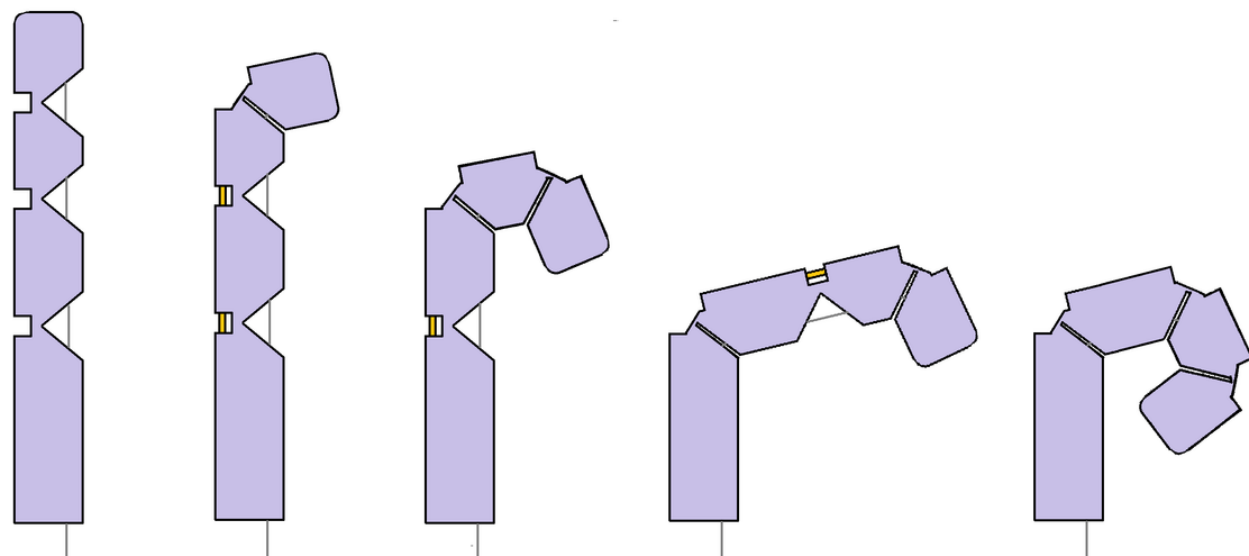
Locking strategies exist for rigid systems, but the integration of an embedded rigid locking element into a soft system is non-existent.

Motivation and Research Gap

The main motivation of the thesis is to make a soft robotic finger that can achieve multiple configurations

How?

By implementing, a discrete, electrically controlled locking mechanism, embedded in a soft finger to hold multiple grasp configurations without sacrificing softness or adding excessive power draw.

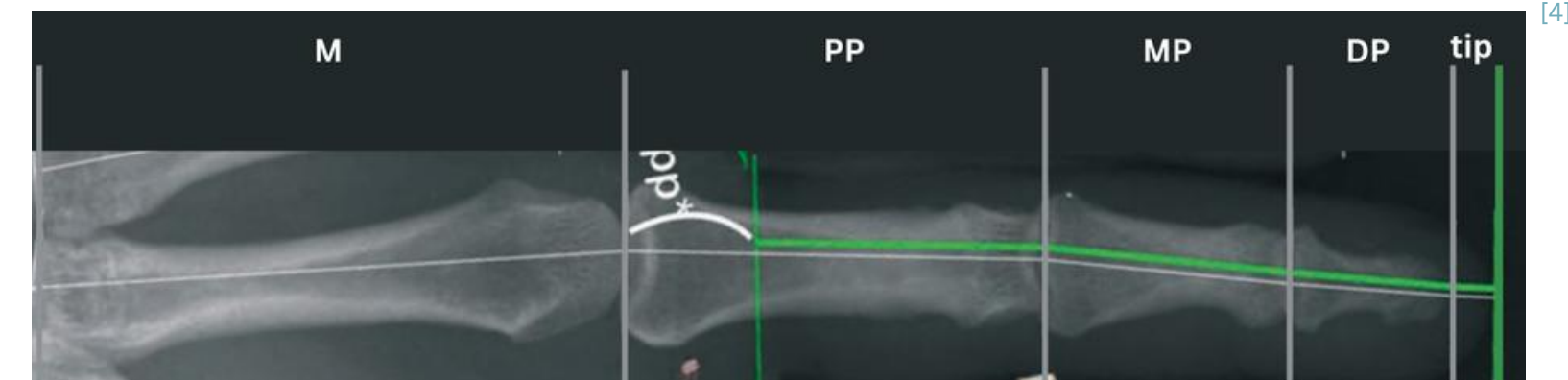
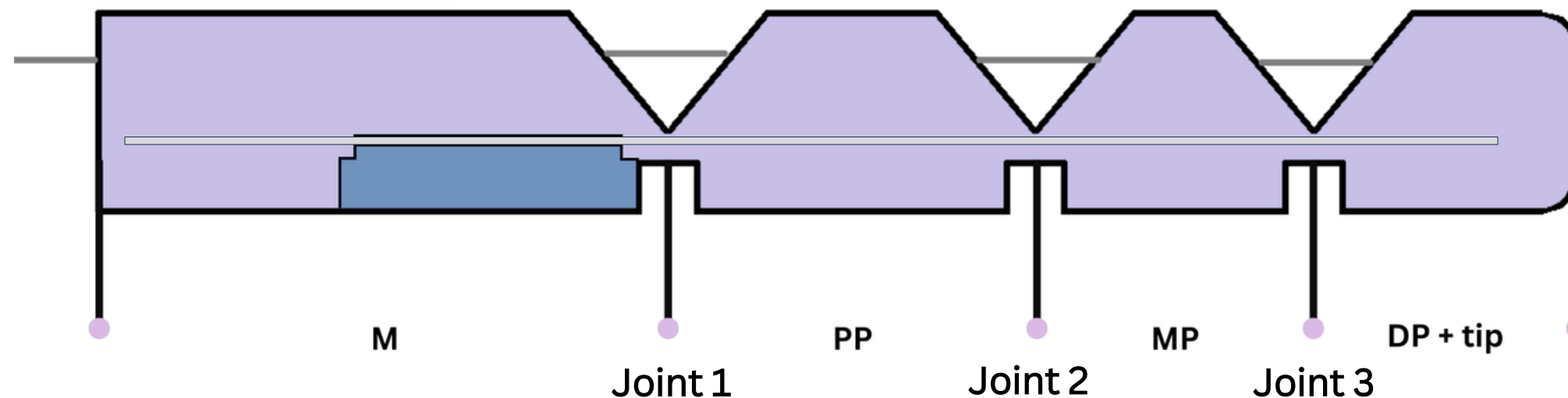


Thesis objective

- Propose and compare multiple candidate mechanisms.
- Select and integrate the locking mechanisms into a silicon-cast, tendon-driven finger.
- Experiment and validate at mechanism- and finger-level

Finger Design

Segment lengths are constrained to within the anatomically derived targets, keeping proportions consistent with the human middle finger.

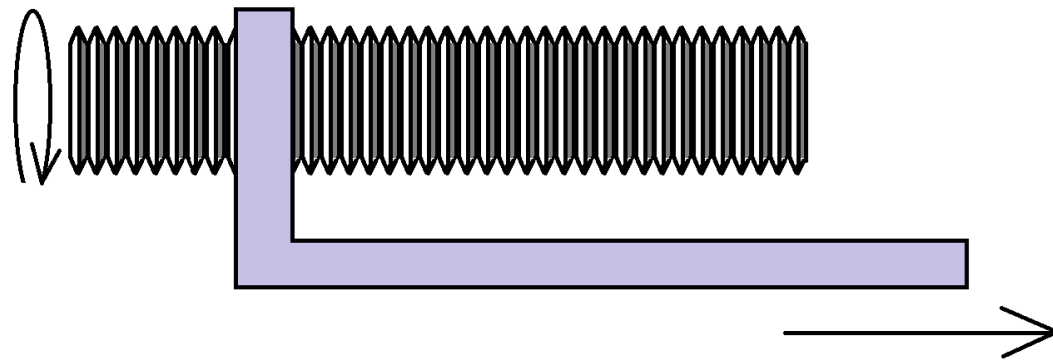


90°
Angle of the finger joint

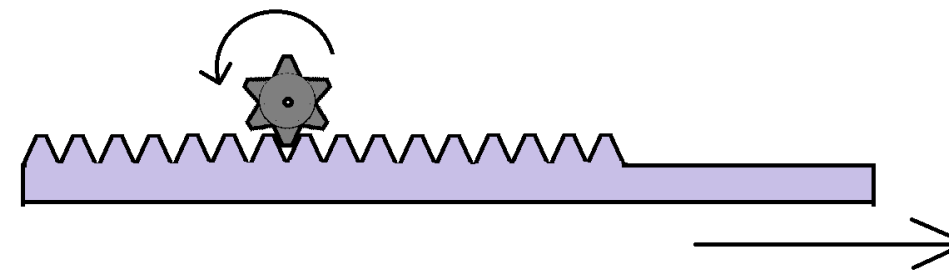
≥250°
Total Range of motion
(Key requirements)

Locking mechanism selection

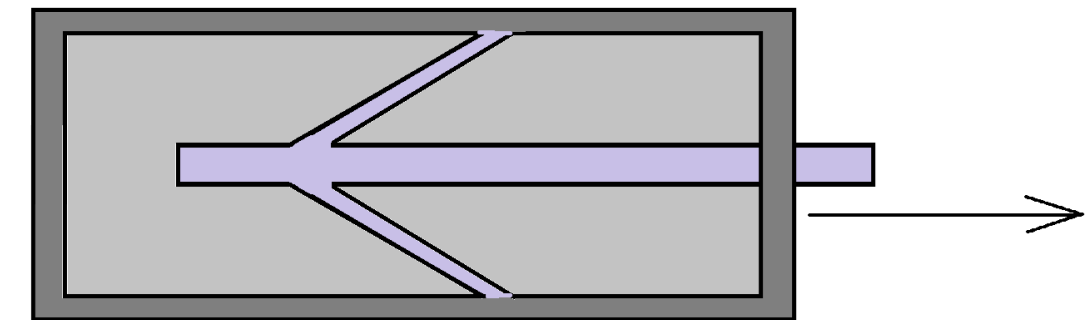
(a) Screw based linear mechanism



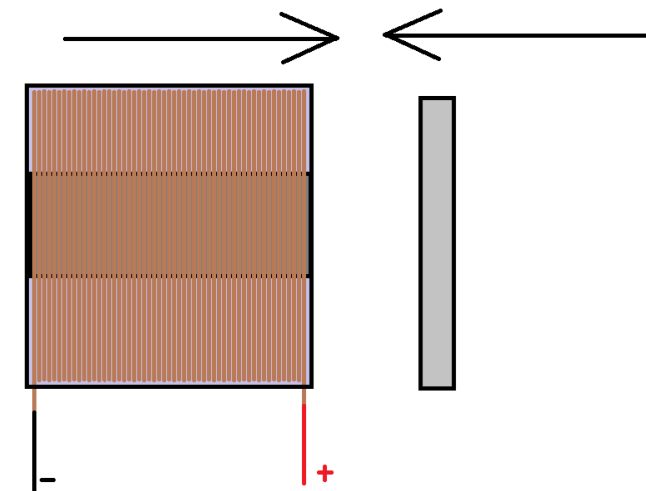
(b) Rack and pinion mechanism



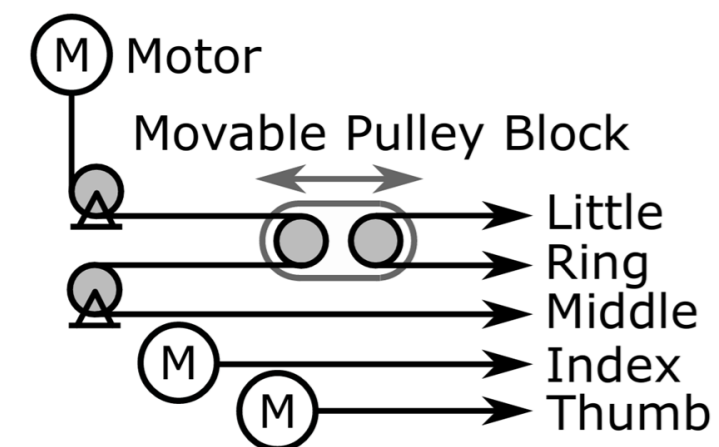
(c) Compliant mechanism



(d) Electromagnetic mechanism



(e) Pulley system



Locking mechanism selection

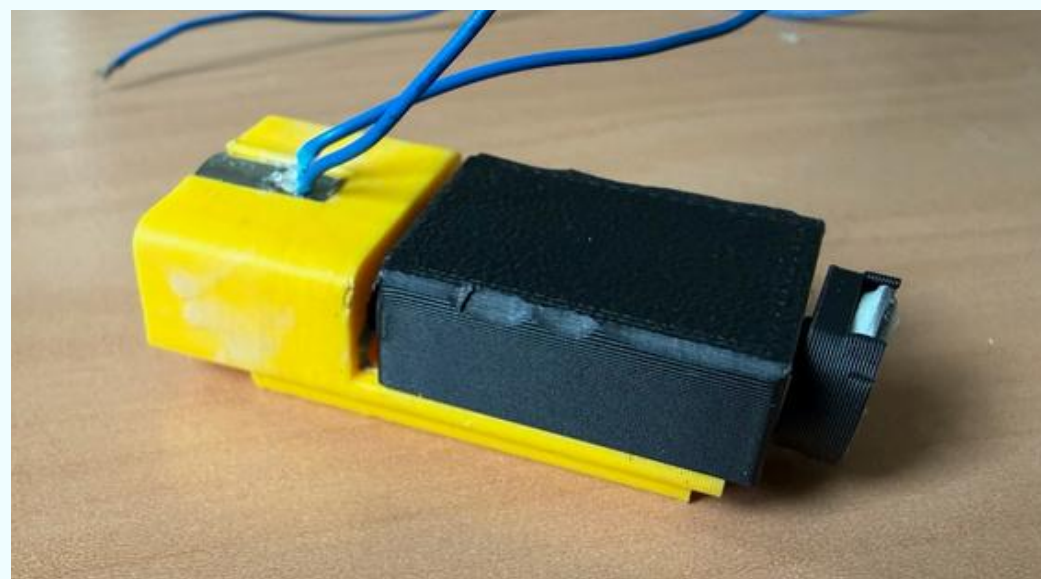
	Rack and pinion (RP)	Magnetic (M)	Compliant (CP)	Screw based Linear (LA)	Pulley (P)
Locking strength (L)	+	+	+	+	+
Energy efficiency (E)	+	-	+	+	+
Size and Availability (S)	+	+	+	+	-
Weight (W)	+	+	+	+	-
Complexity (C)	0	+	-	+	-
Compatibility with soft robotics (R)	+	+	+	+	-
Durability(D)	+	+	-	+	+
Final score	+	+	0	+	-

Selected: Linear Actuator (LA) and Magnetic mechanisms advanced to prototyping; Compliant and Rack-and-Pinion were built then discontinued; Pulley was eliminated on paper

Mechanism designs

Electromagnet Mechanism

- This mechanism relies on an electromagnet and the metal piece for locking.
- An electromagnet rated at 24 V, 1.0W, is used for this purpose.



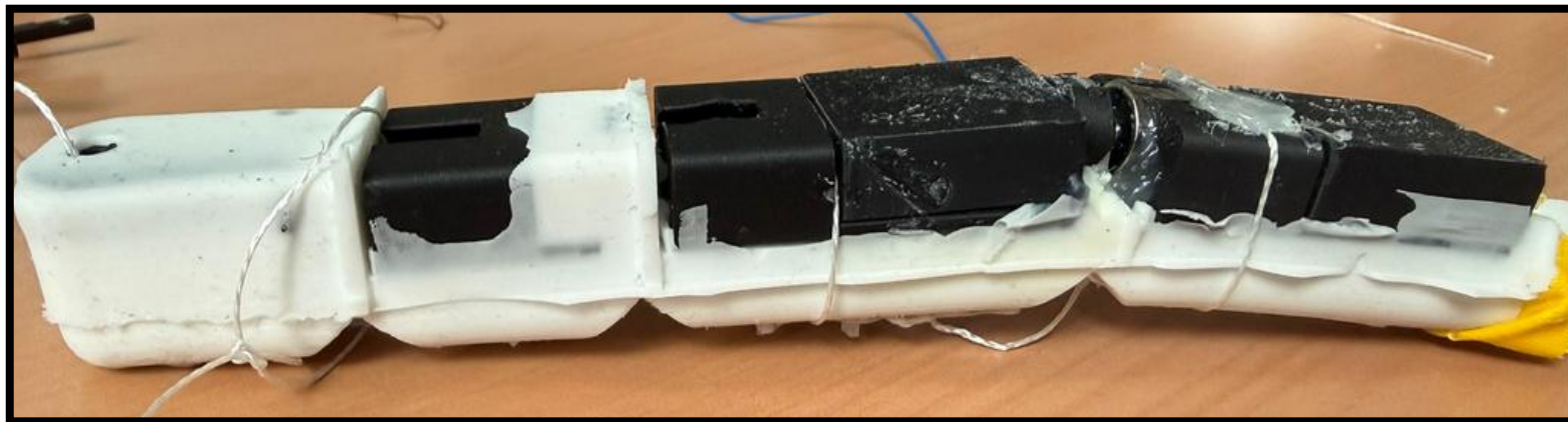
Linear actuator Mechanism

- CA-1006 micro stepper motor - 9 mm stroke is used for this mechanism
- Screw-based threaded shaft drives non-backdrivable linear lock.

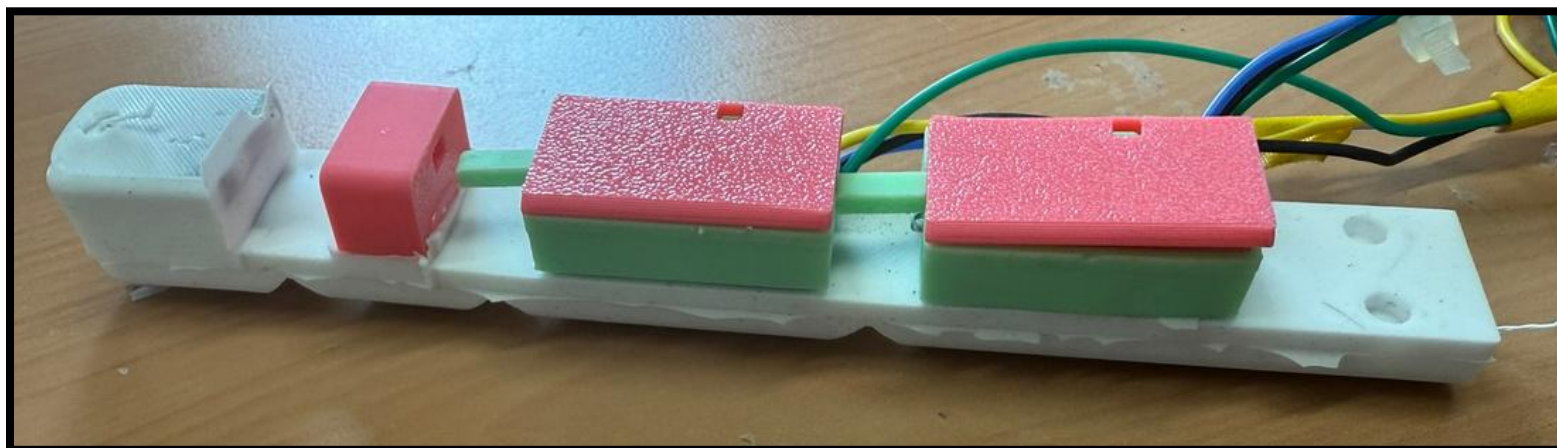


Final fingers

With the electromagnet mechanisms



With the linear actuator mechanisms



- Two of these final fingers are tested to find the best mechanism, and the best candidate can be used for further construction.
- Locking currently implemented at Joints 1 and 2
- Tendon-driven actuation with PTFE-lined channels
- Anatomically-proportioned segments (166 mm total length)

Mechanism-Level Experiments

Activation time

Mechanism	Phase	Mean (s)	Std. Dev. (s)
LA	Locking	5.56	0.29
	Unlocking	5.04	0.10
EM	Locking	2.25	0.05
	Unlocking	0.00	0.00
EM (Uncharged)	Locking	54.42	-
	Unlocking	0.00	-

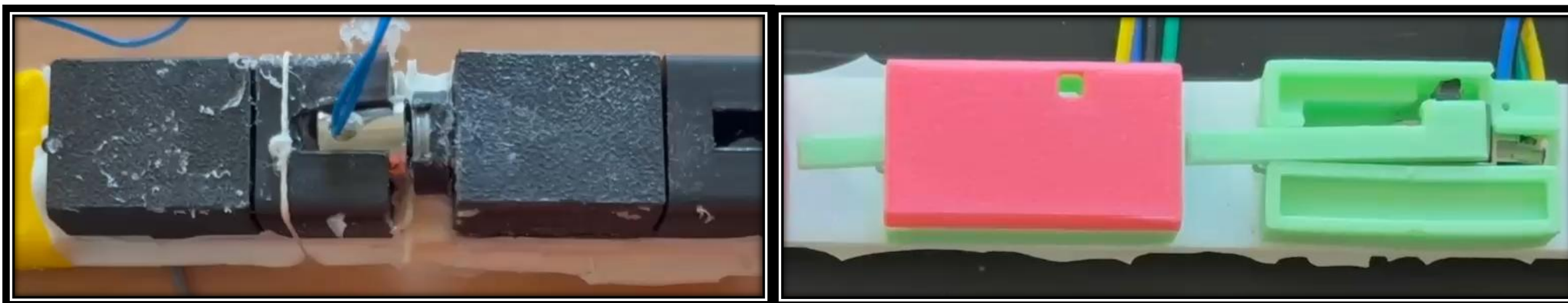
Avg Locking time

5.5 s

Linear actuator

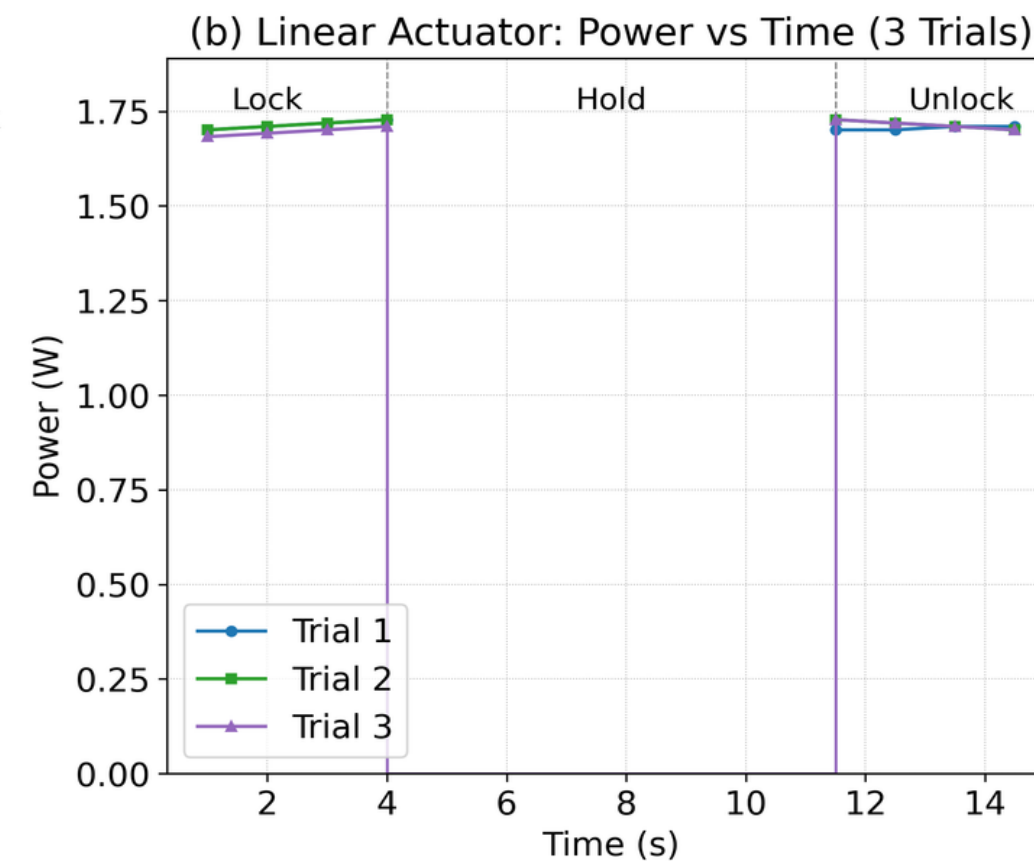
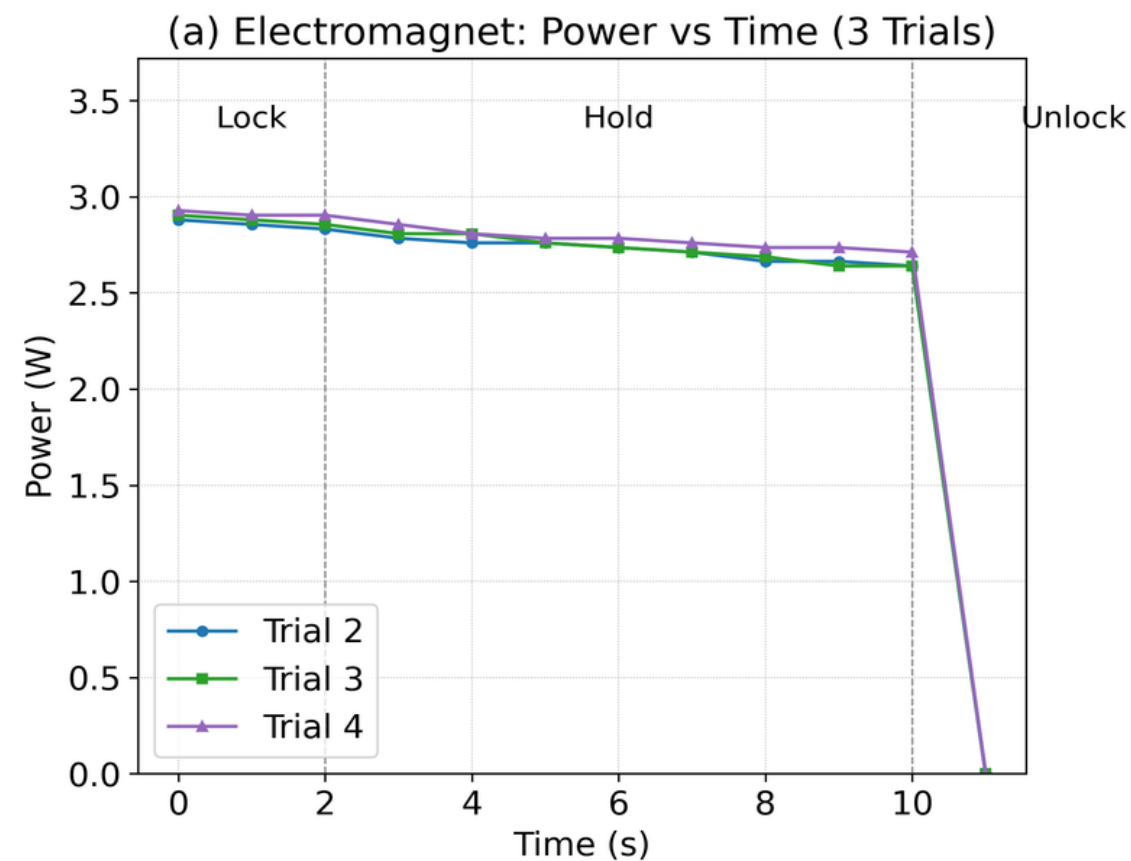
2.5 s

Electromagnet



Power consumption

Mechanisms	Voltage (V)	Mean Power Locking (W)	Mean Power Holding (W)	Mean Power Unlocking (W)	Energy/Cycle (J)
LA	4.5	1.7085	0.0000	1.7115	13.680
EM	24	2.8827	2.7370	0.0000	30.544
EM (Uncharged)	24	2.9333	2.7000	0.0000	211.128



Avg Locking Power

1.71 W

Linear actuator

2.88 W

Electromagnet

Load carried

Mechanisms	Bending Angle (°)	Cumulative Load (N)	Maximum Load (N)
LA	12	1.726	8.25
EM	30	0.883	0.883

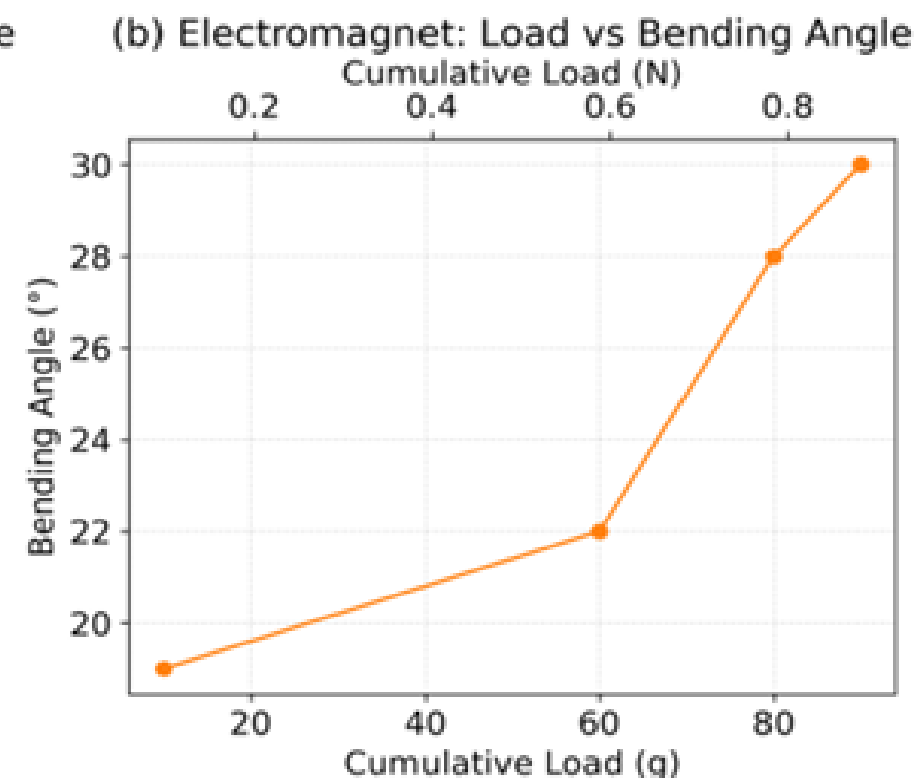
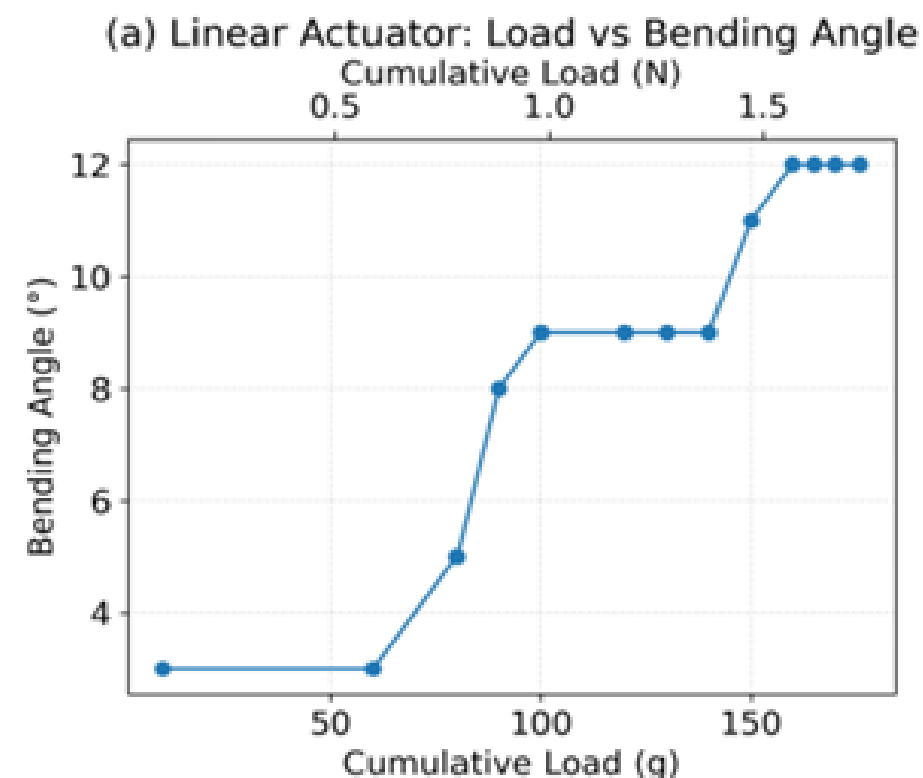
Maximum Load

8.25 N

Linear actuator

0.88 N

Electromagnet



Mechanisms overview

Criteria	Linear actuator	Electromagnet	Preferred
Avg. locking time	5.5 s	2.5 s	Electromagnet
Avg. locking power	1.74W	2.88 W	Linear actuator
Max. load Applied	8.25 N	0.88 N	Linear actuator
		Final Choice	Linear actuator

Finger-Level Experiments

Tendon pull force

- To determine the force required to pull the tendon of the finger through its full range of motion.
- This force is used to calculate the torque required for the actuating motor.

$2.78 \pm 0.24\text{N}$

Total pull force required

Finger tip strength

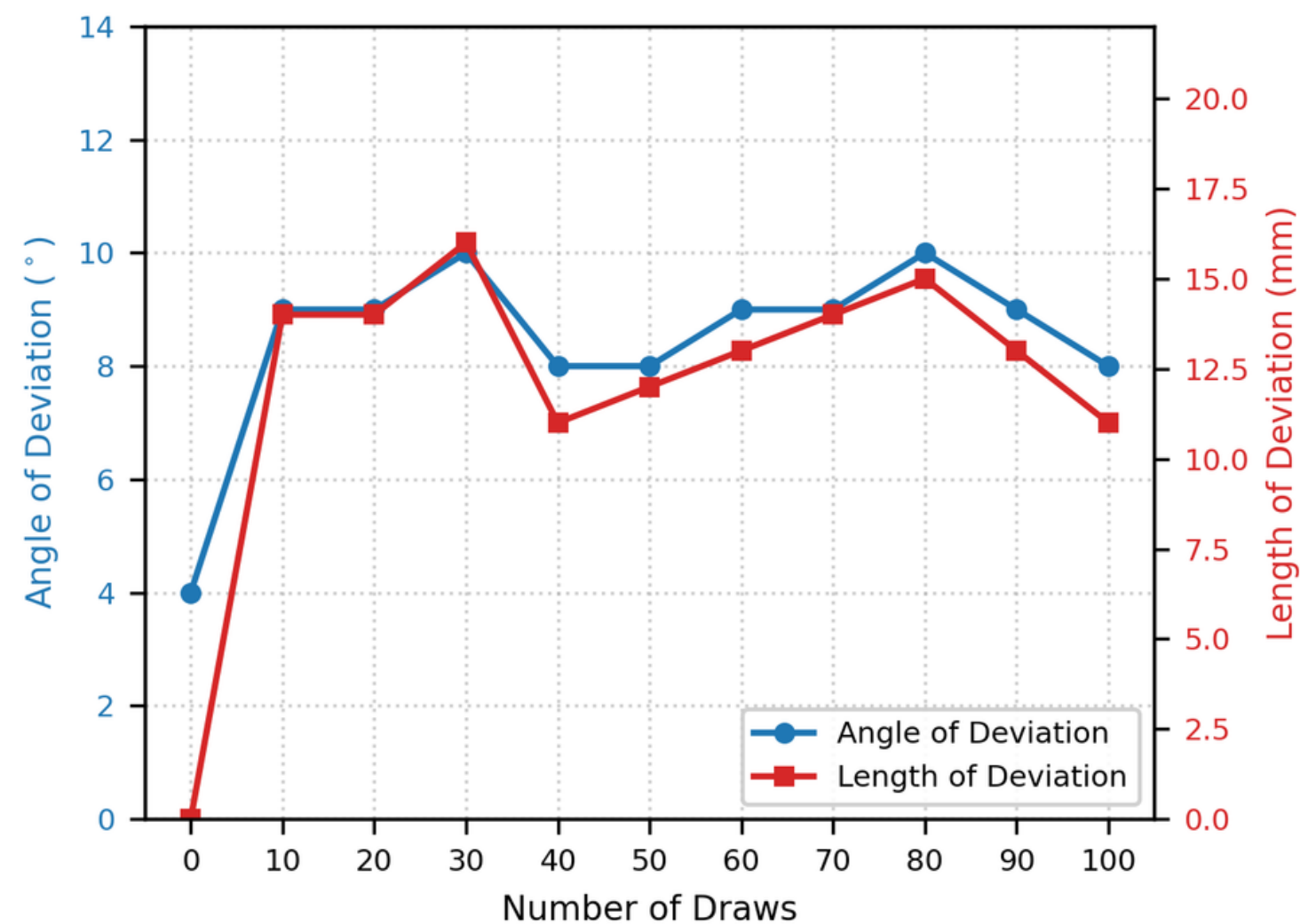
- To measure the grip strength of the finger, quantified as the force exerted by the fingertip when fully actuated.
- Key indicator of the finger's ability to grasp and hold objects during practical use.

$4.1 \pm 0.5\text{ N}$

Finger tip grip strength

Precision and repeatability

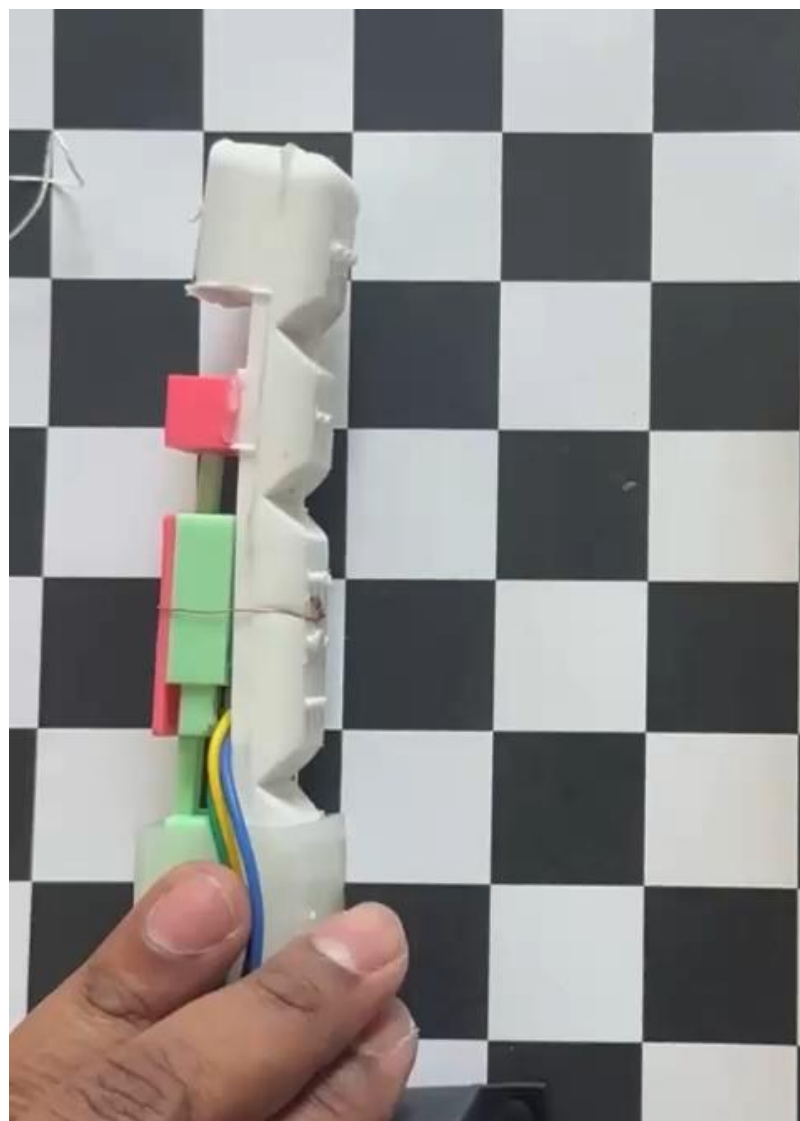
- To evaluate the finger's precision and repeatability by measuring the fingertip's deviation from its initial position over 100 repeated actuation cycles (draws).
- Critical for evaluating the long-term reliability, under cyclic loading.



10°
Angle of max
deviation

Range of motion

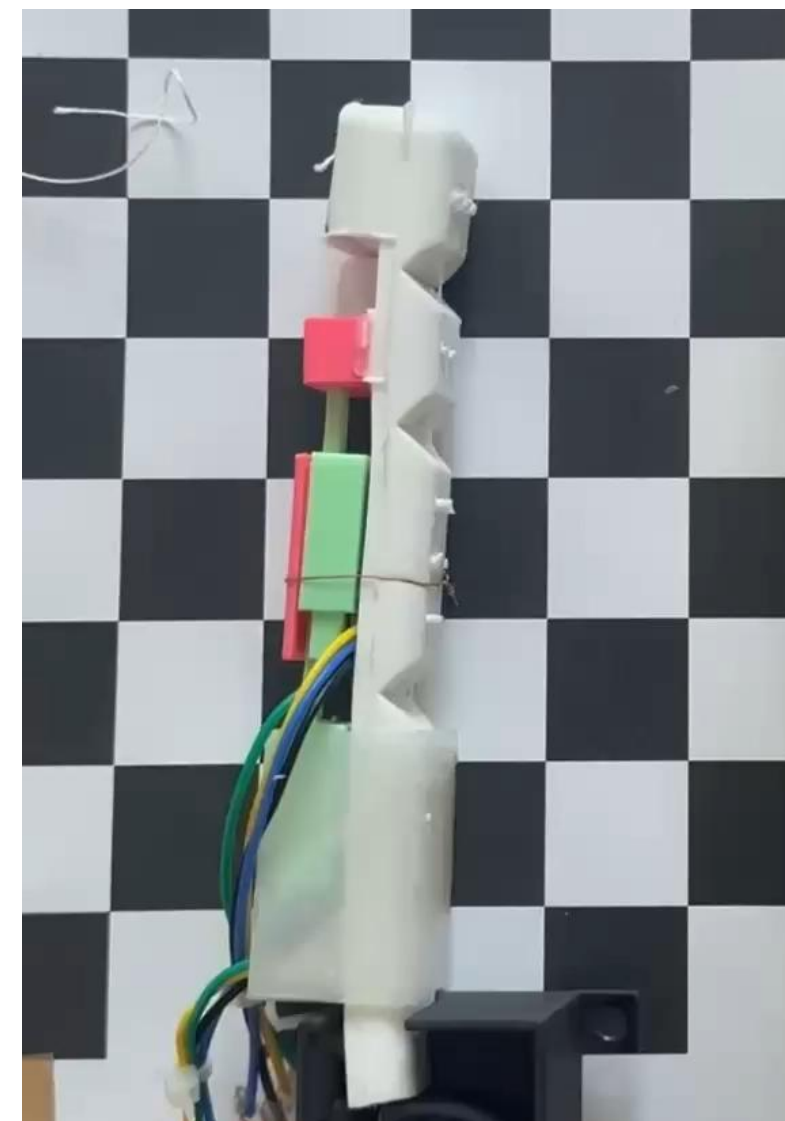
Joint 1 Locked



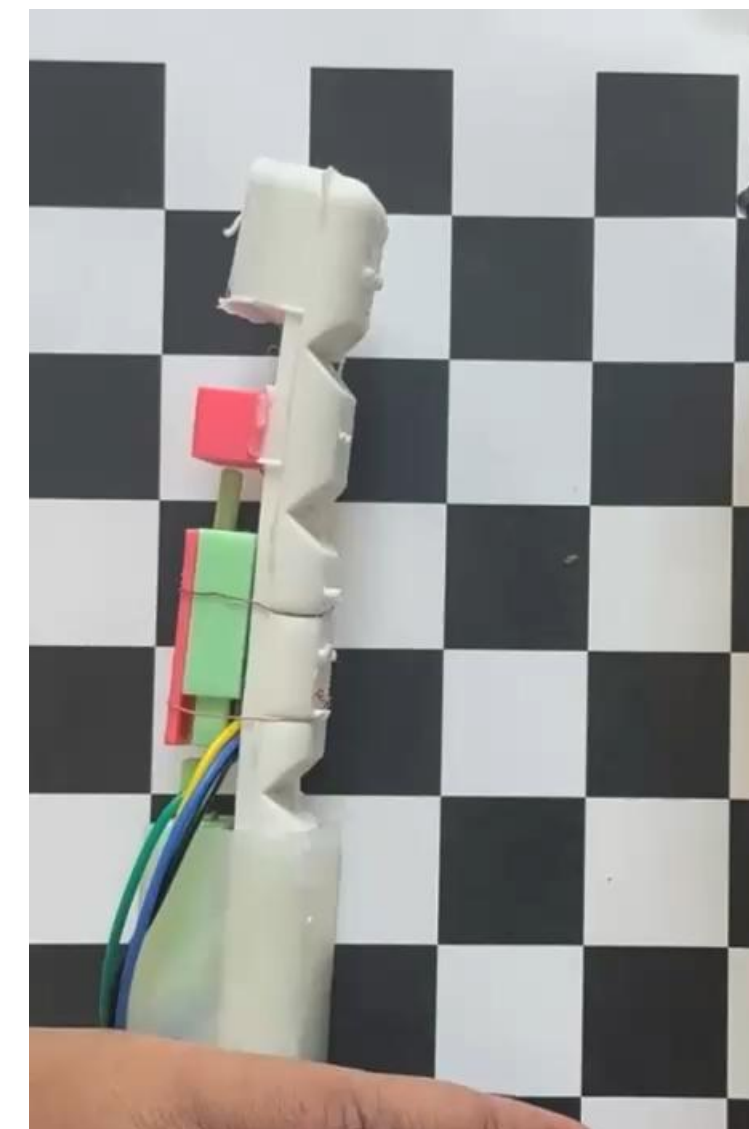
Joint 2 Locked



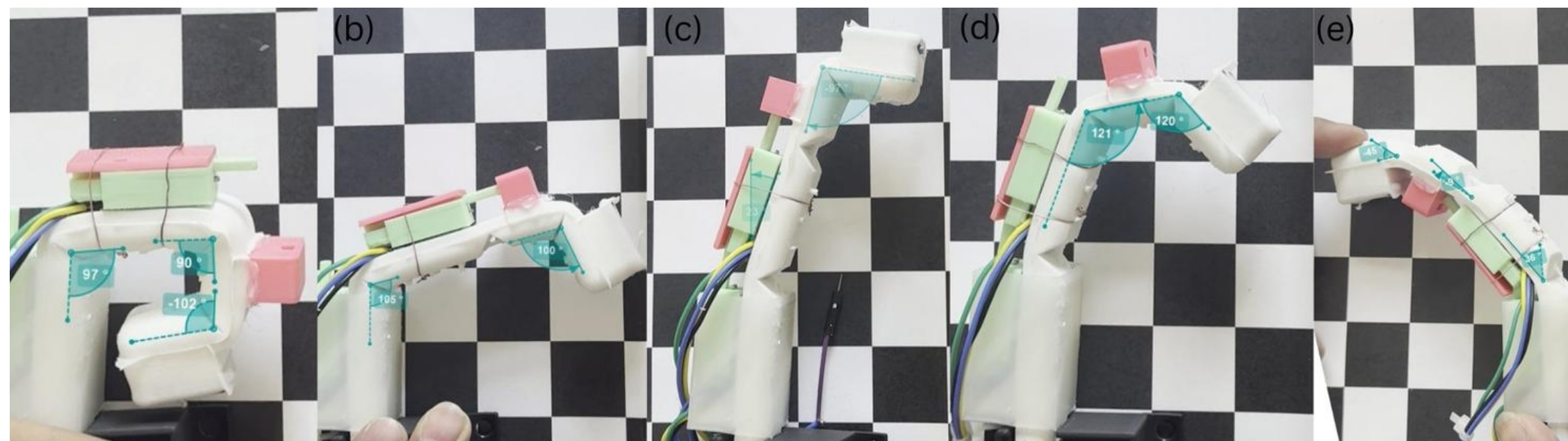
Joint 1 and 2 Locked



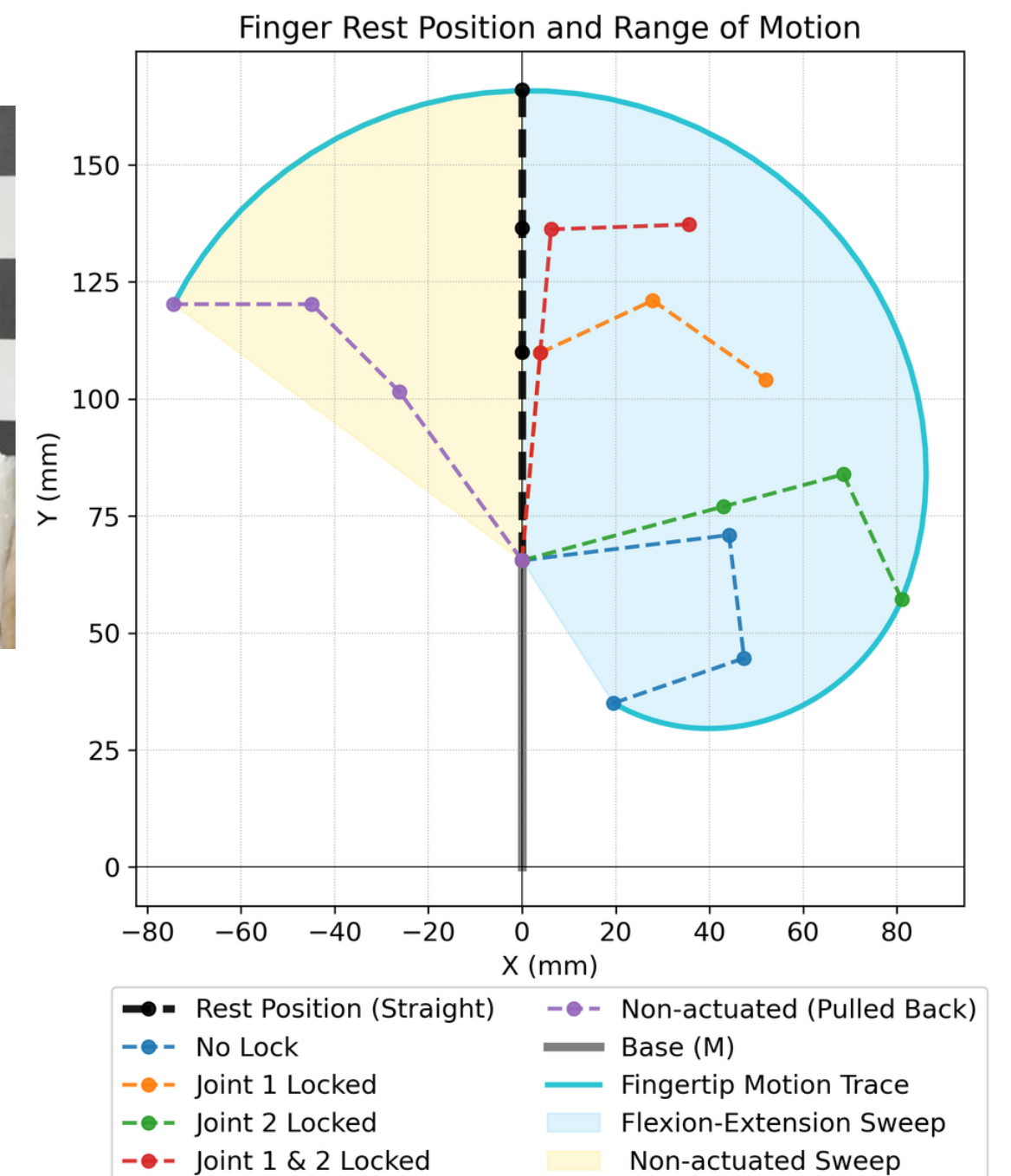
No Joint Locked



Range of motion



Configuration	Joint 1 (°)	Joint 2 (°)	Joint 3 (°)	Total (°)
No lock	83	90	78	251
Joint 1 locked	5	60	60	125
Joint 2 locked	75	0	80	155
Joint 1 and 2 locked	5	0	83	110
Non-actuated (pulled backward)	-36	-9	-45	-90



Overview

Requirement	Target	Achieved (LA Finger)	
Locking time	$\leq 5 \text{ s}$	5 s	✓
Holding load without failure	$\geq 2 \text{ N}$	8.25 N	✓
Mechanism power draw	$\leq 5 \text{ W}$	1.71 W	✓
Grip strength	$\geq 3 \text{ N}$	4.1 N	✓
Fingertip repeatability	$\leq 10^\circ$	10°	✓
Cumulative range of motion	$\geq 250^\circ$	Confirmed across configs.	✓
Mass		66 g	

The target values are taken from the chapter 3 of the thesis.

Conclusion

- Soft finger with an embedded locking mechanism was **designed, fabricated, and validated**.
- Candidate mechanisms were compared then the **linear actuator** and **electromagnet** were carried through to the final designs.
- These two mechanisms were put through an experimental analysis, where the **linear actuator** was chosen as the final candidate mechanism.
- Experimental results confirm posture retention **under load** without compromising its inherent **soft properties**.



Future works

Motor actuation

Use measured tendon-pull-force data to calculate the required motor torque for optimal actuator selection.

Full Hand

Scale from a single finger to five fingers integrated into a complete soft robotic hand.



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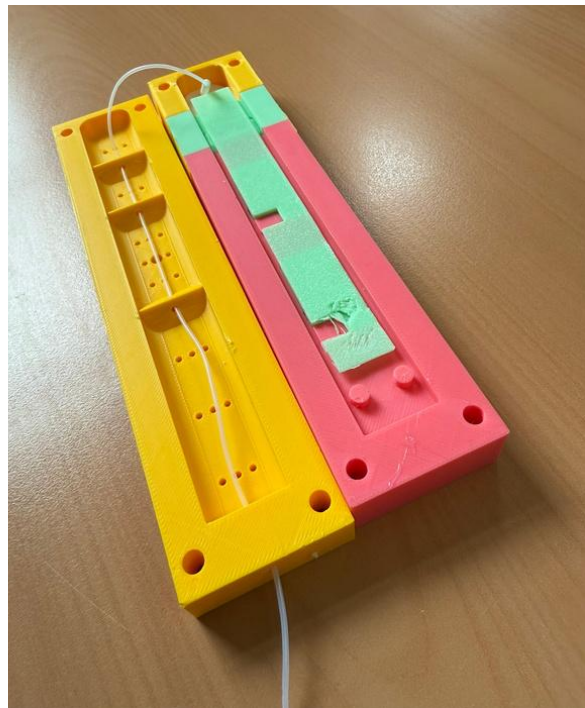
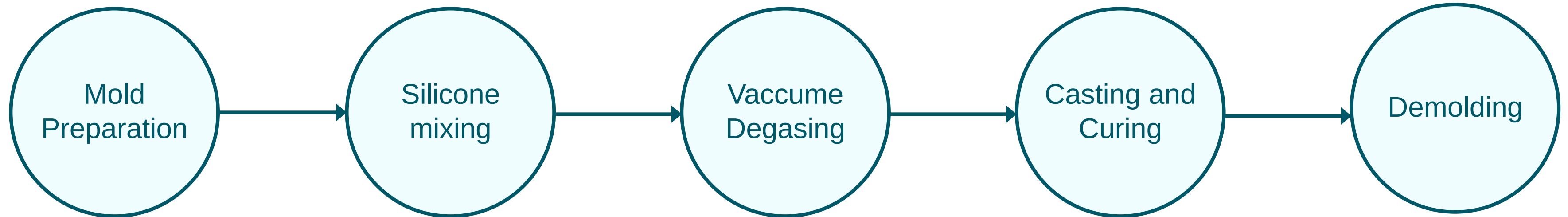
Thank you for your attention

Any
questions?

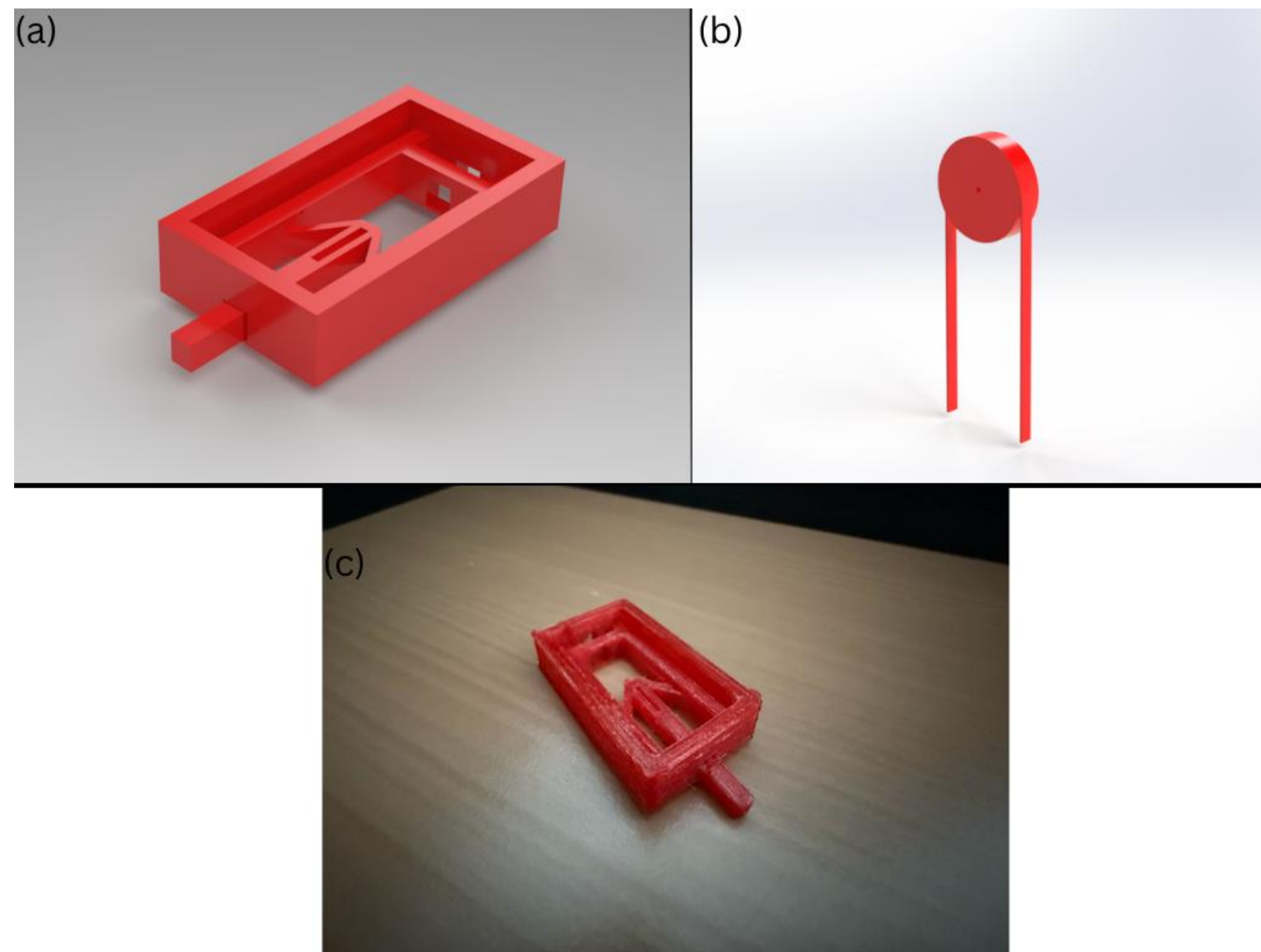
References

1. Peerdeman,B et al., UT hand I:A lock-based underactuated hand prosthesis. In: Mechanism and Machine Theory.
2. Pagoli,A et al., A Soft Robotic Gripper With an Active Palm and Reconfigurable Fingers for Fully Dexterous In-Hand Manipulation. In: IEEE Robotics and Automation Letters.
3. Bolanakis,G.and Papadopoulos,E. Introducing Mag-Nets: Rapidly Bending Electromagnetic Actuators for Self-Contained Soft Robots. In: IEEE Robotics and Automation Letters.
4. Buryanov,A and Kotiuk,V. Proportions of Hand Segments. In: International Journal of Morphology.

Finger fabrication

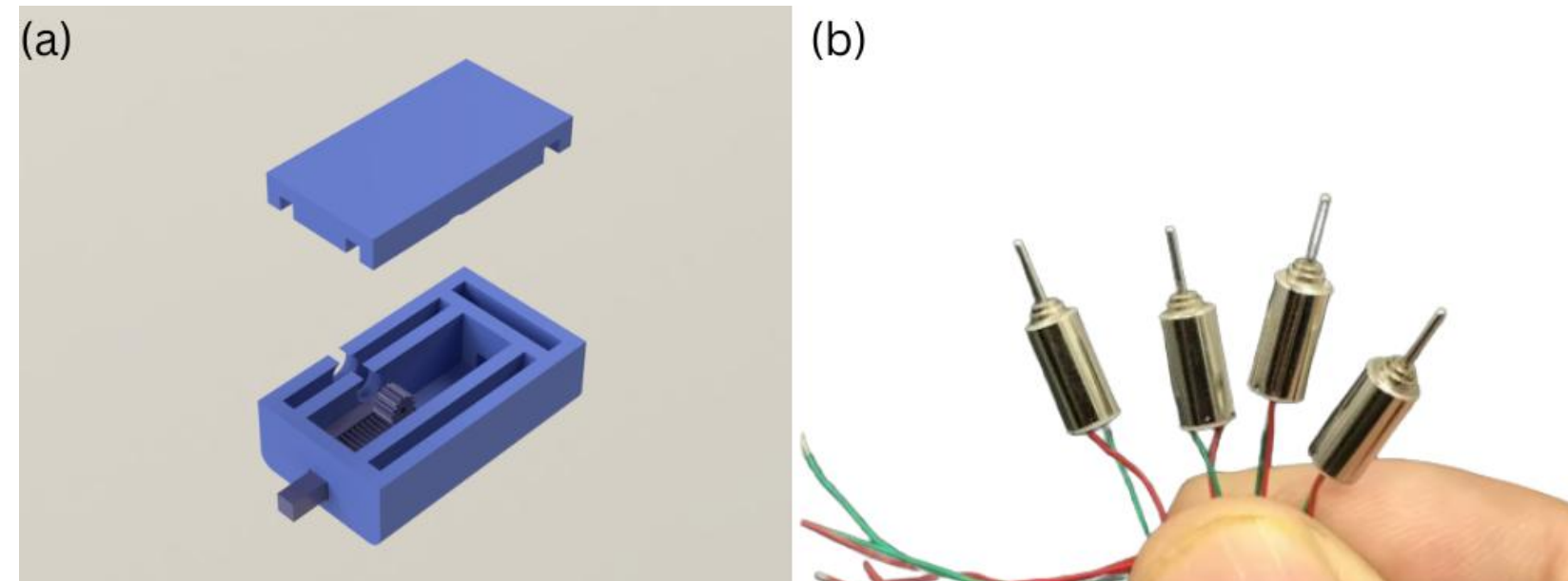


Compliant mechanism

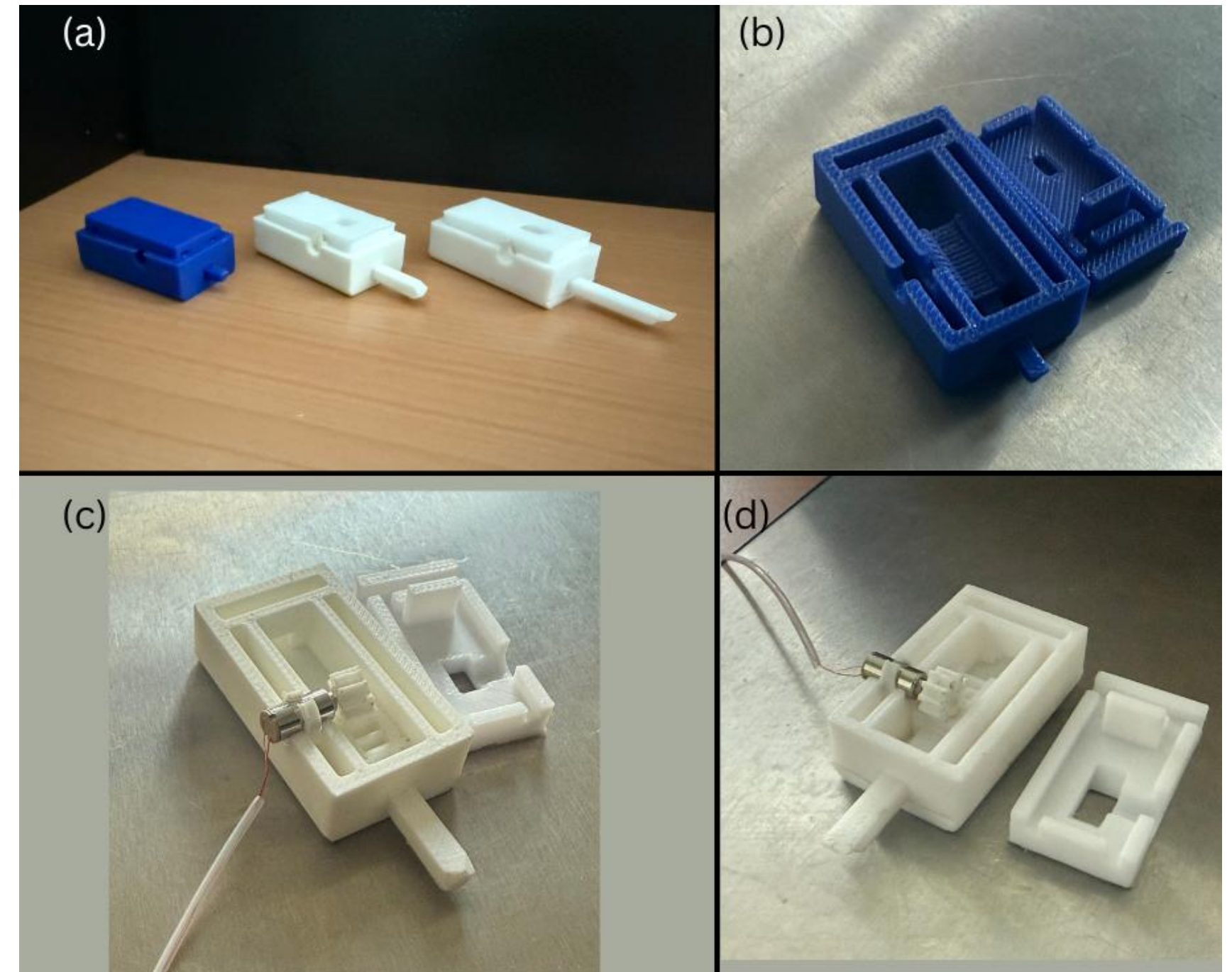


Rack and pinion mechanism

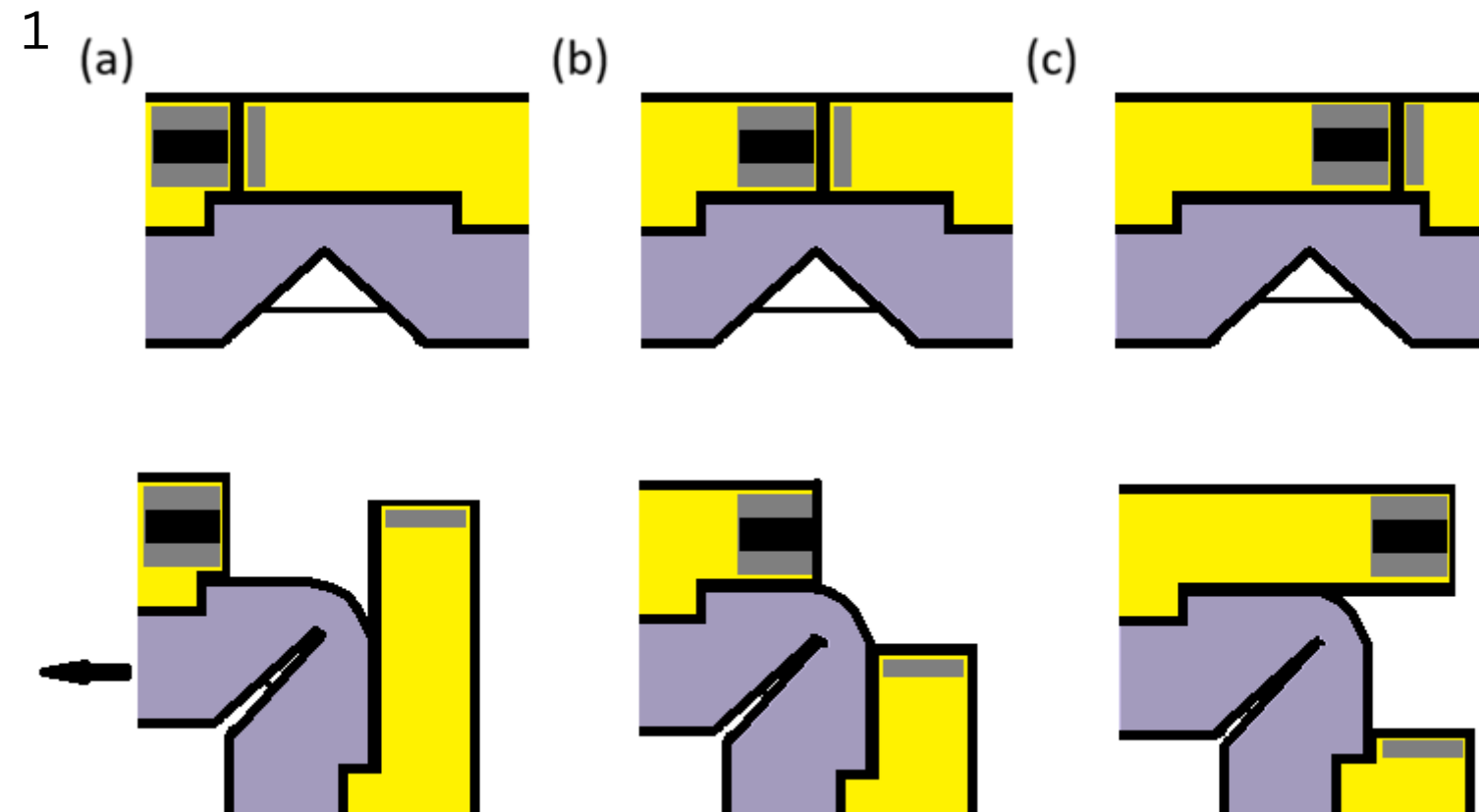
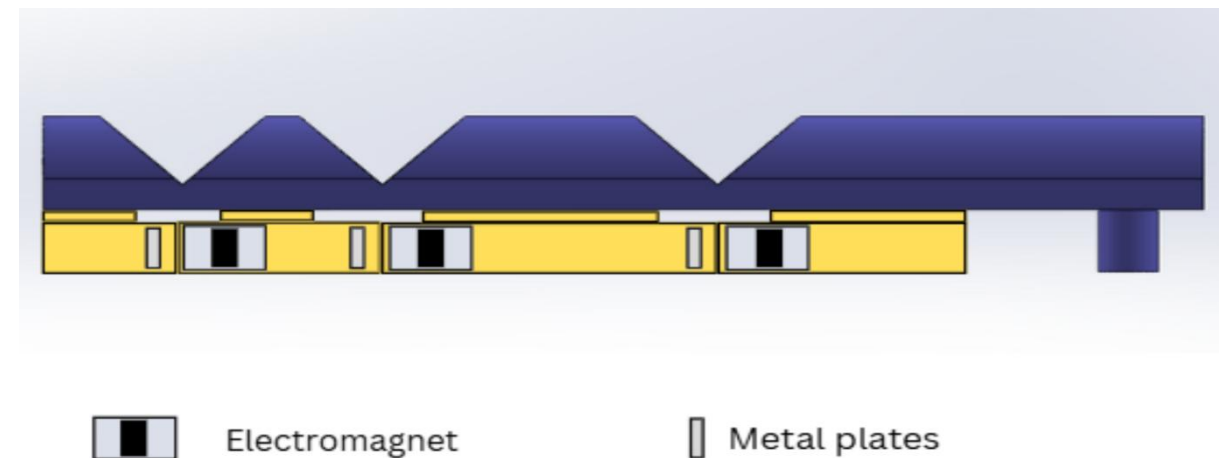
1



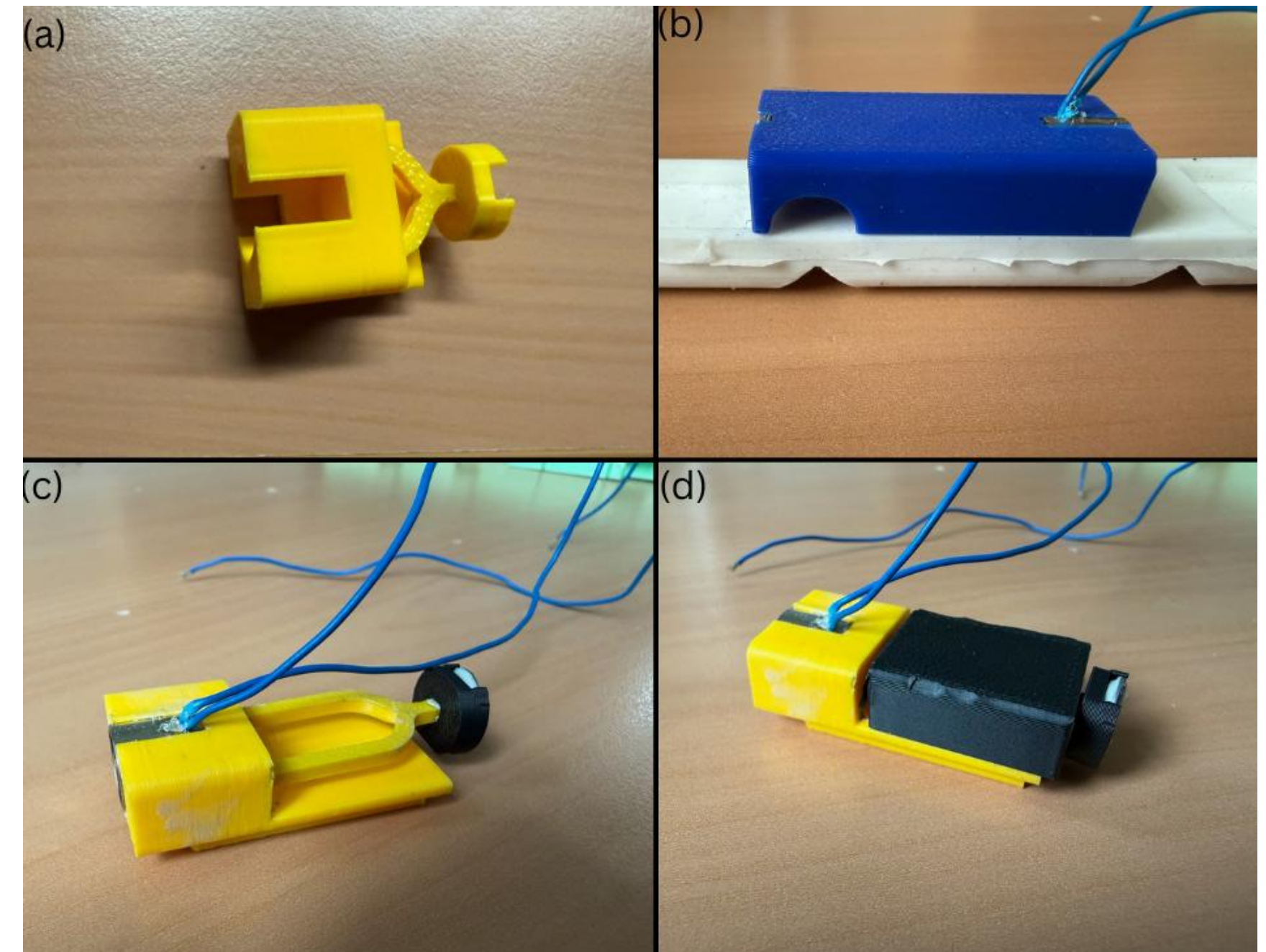
2



Electromagnet mechanism

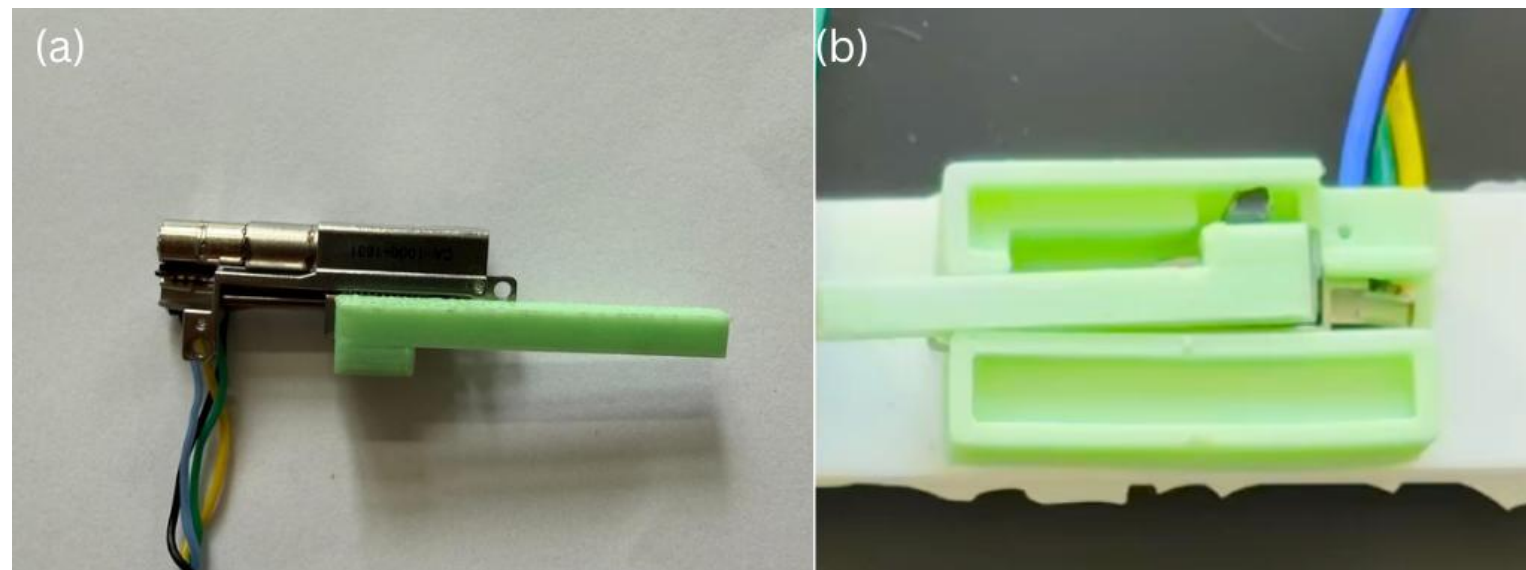


2

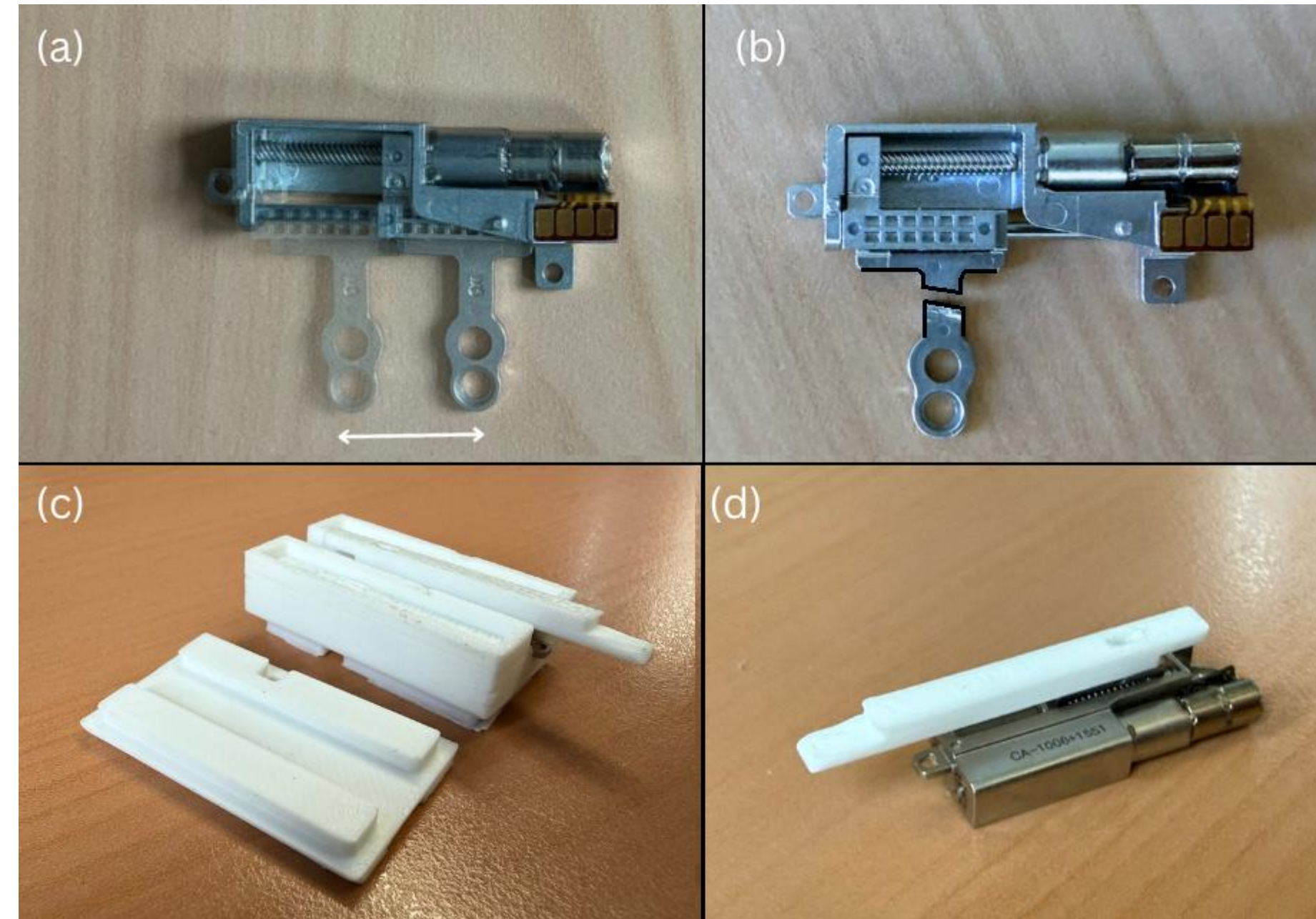


Linear actuator mechanism

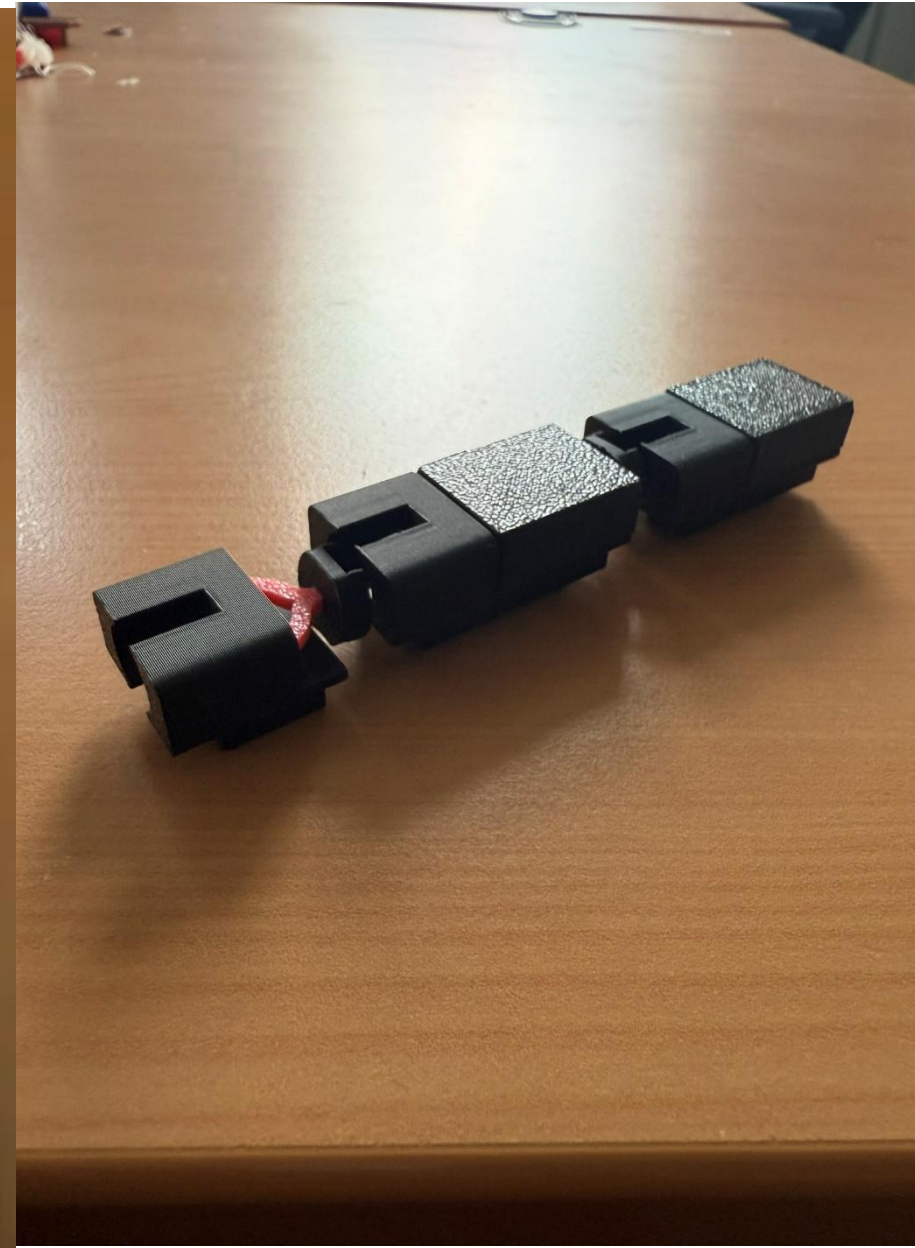
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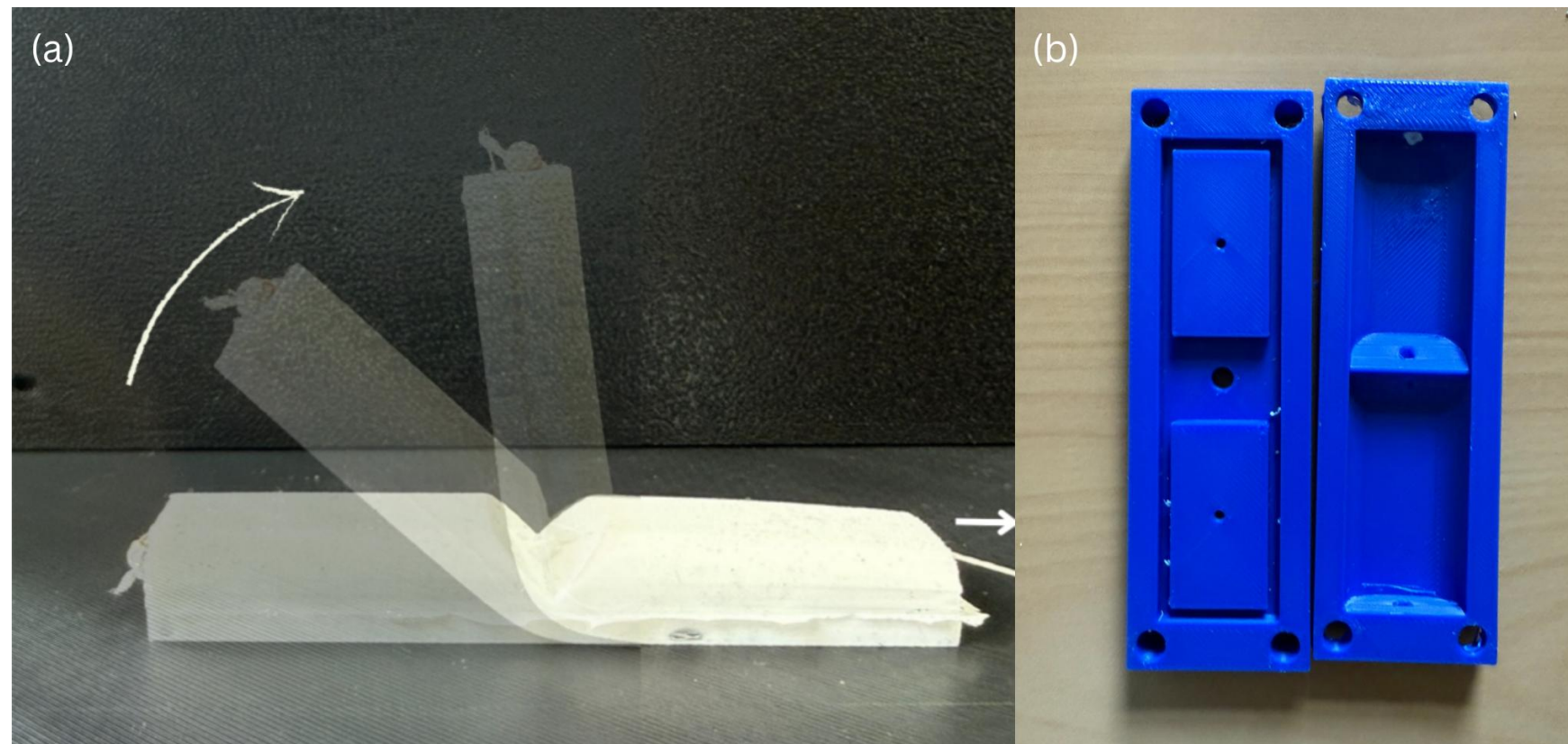
2



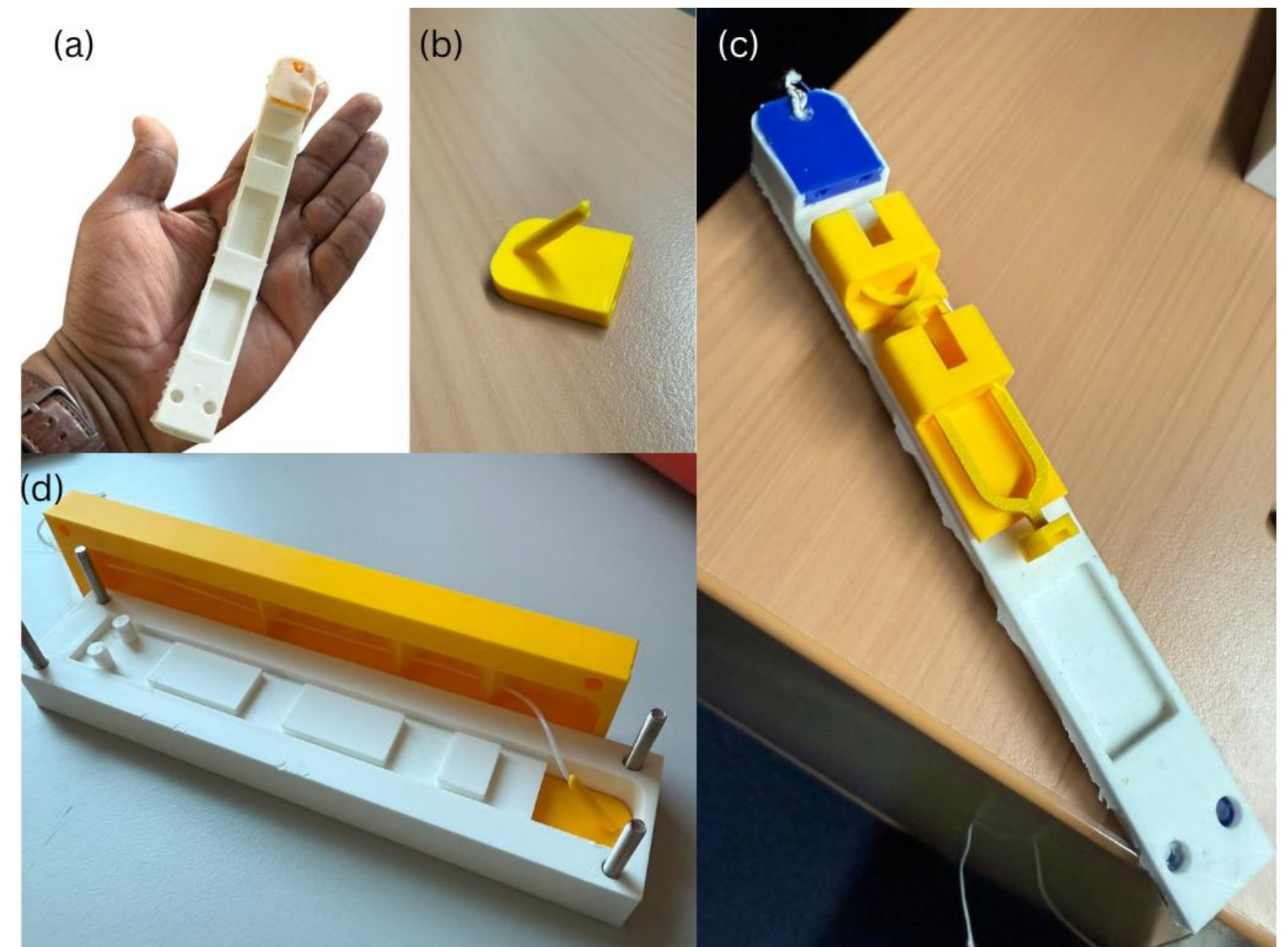
Spring Steel and PLA backbone



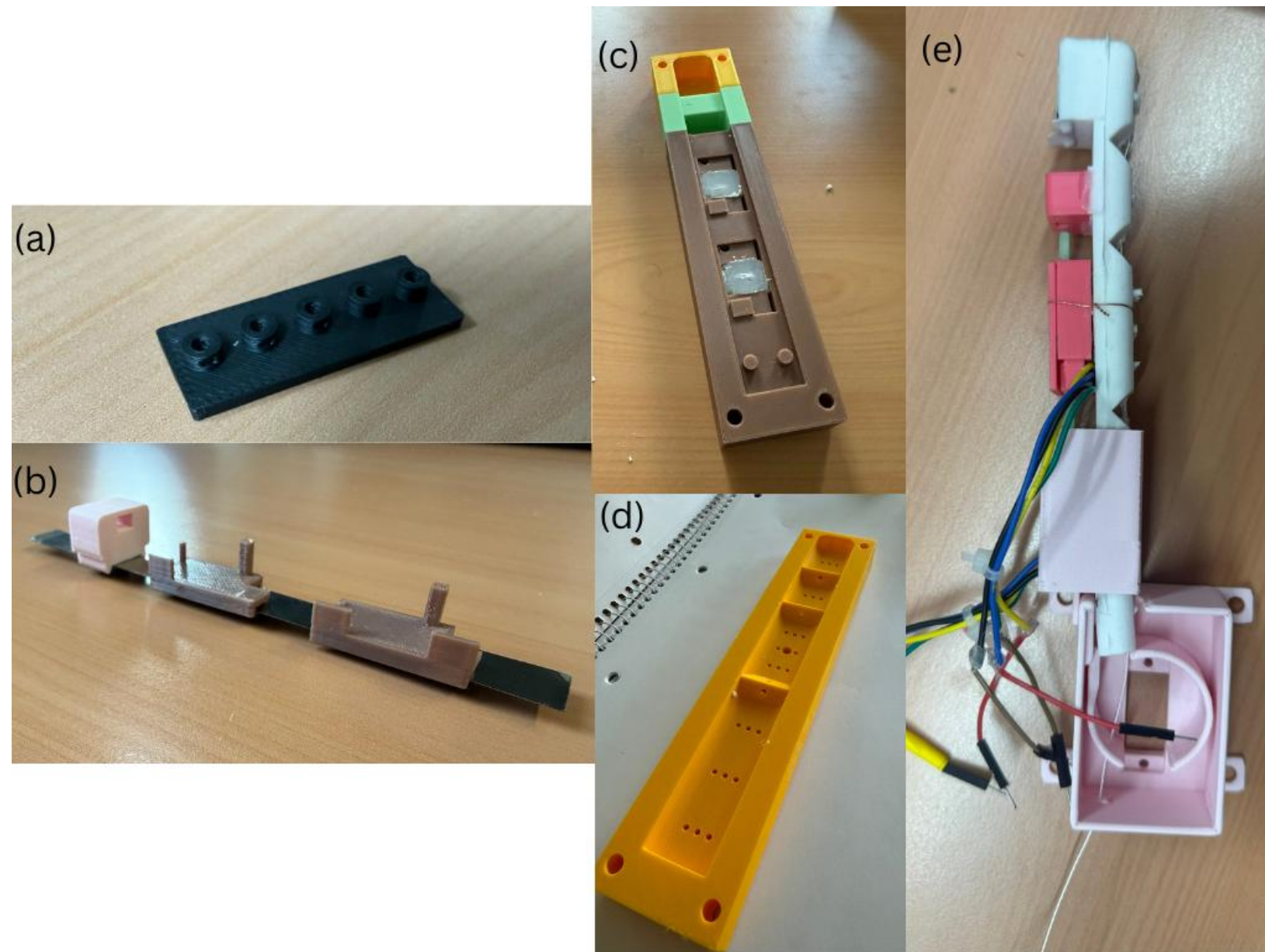
Flexion motion



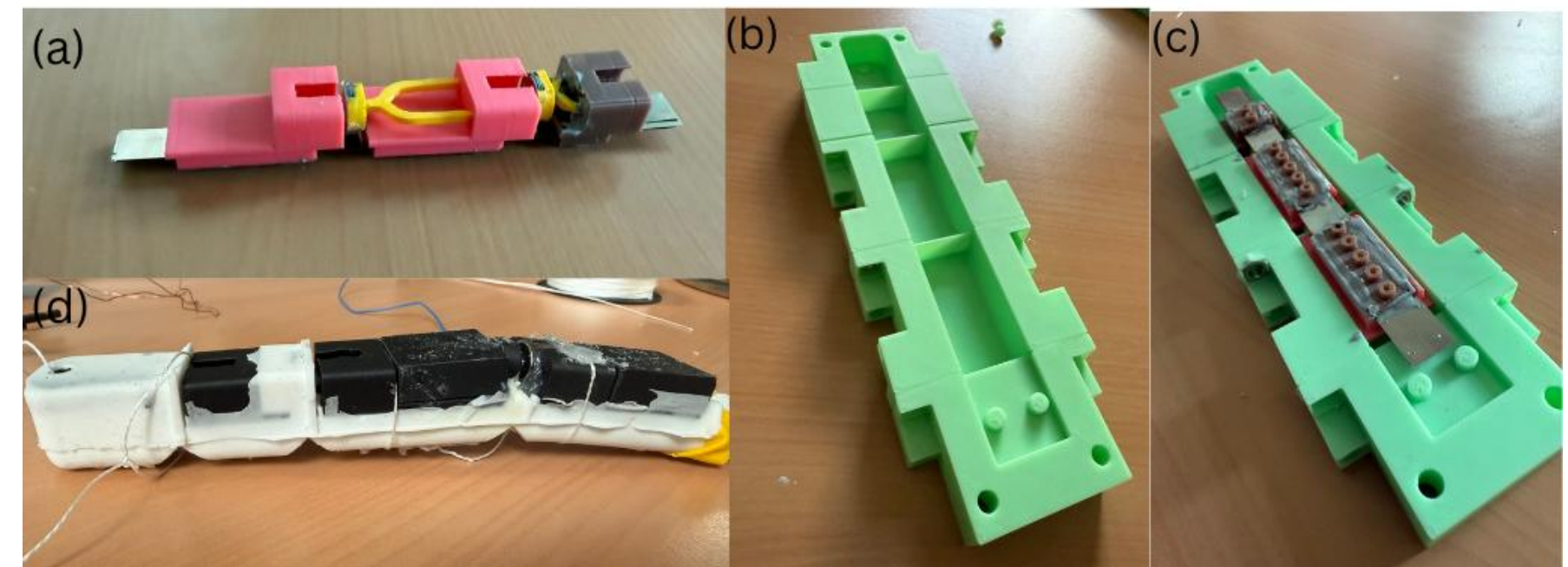
Modular finger



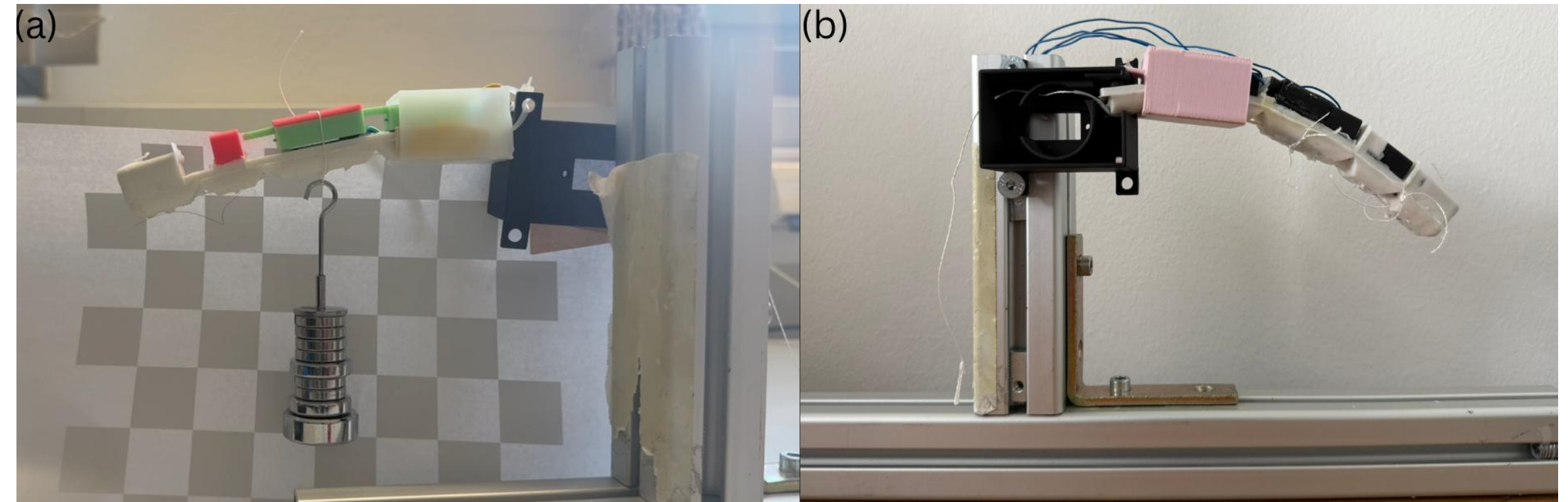
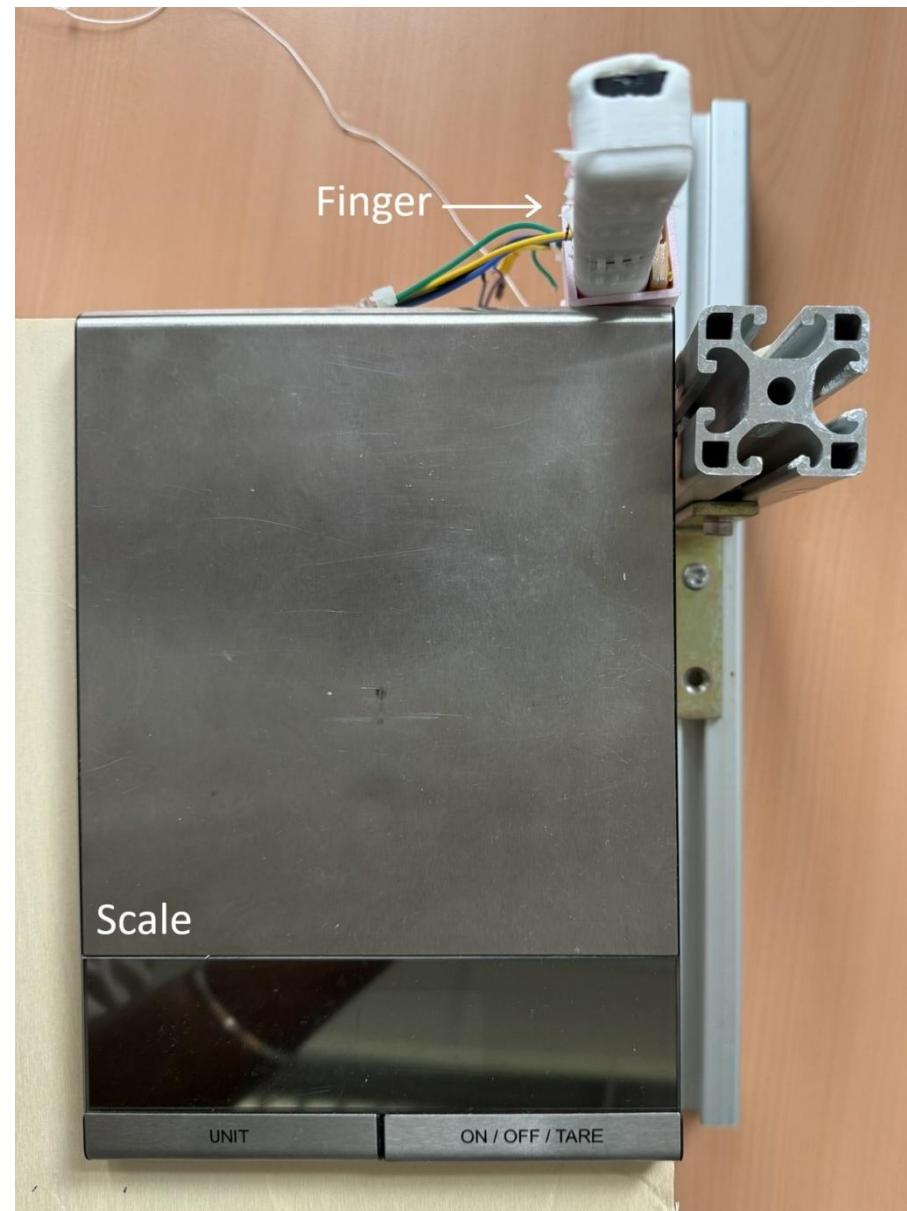
LA finger



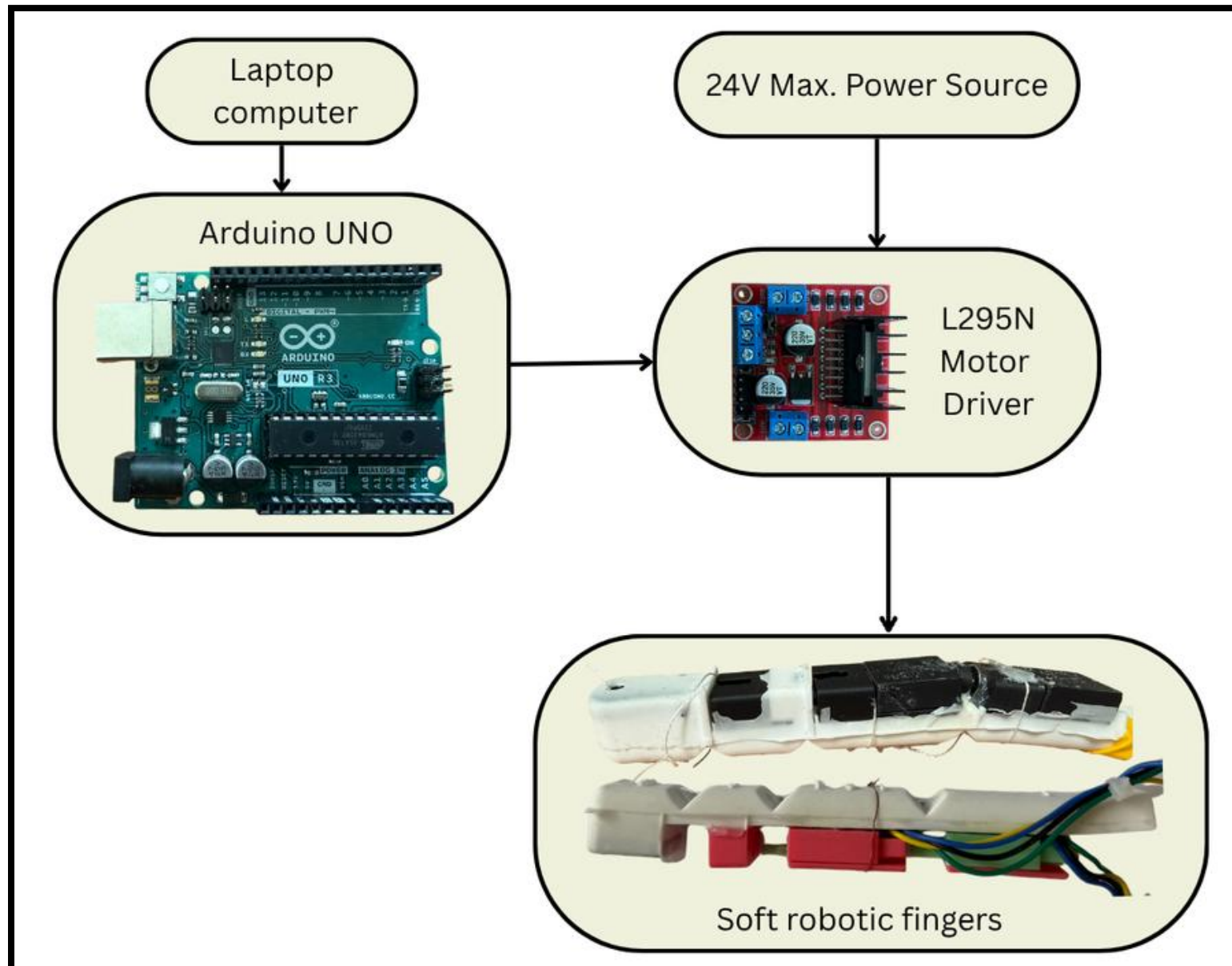
EM finger



Experiments



Control

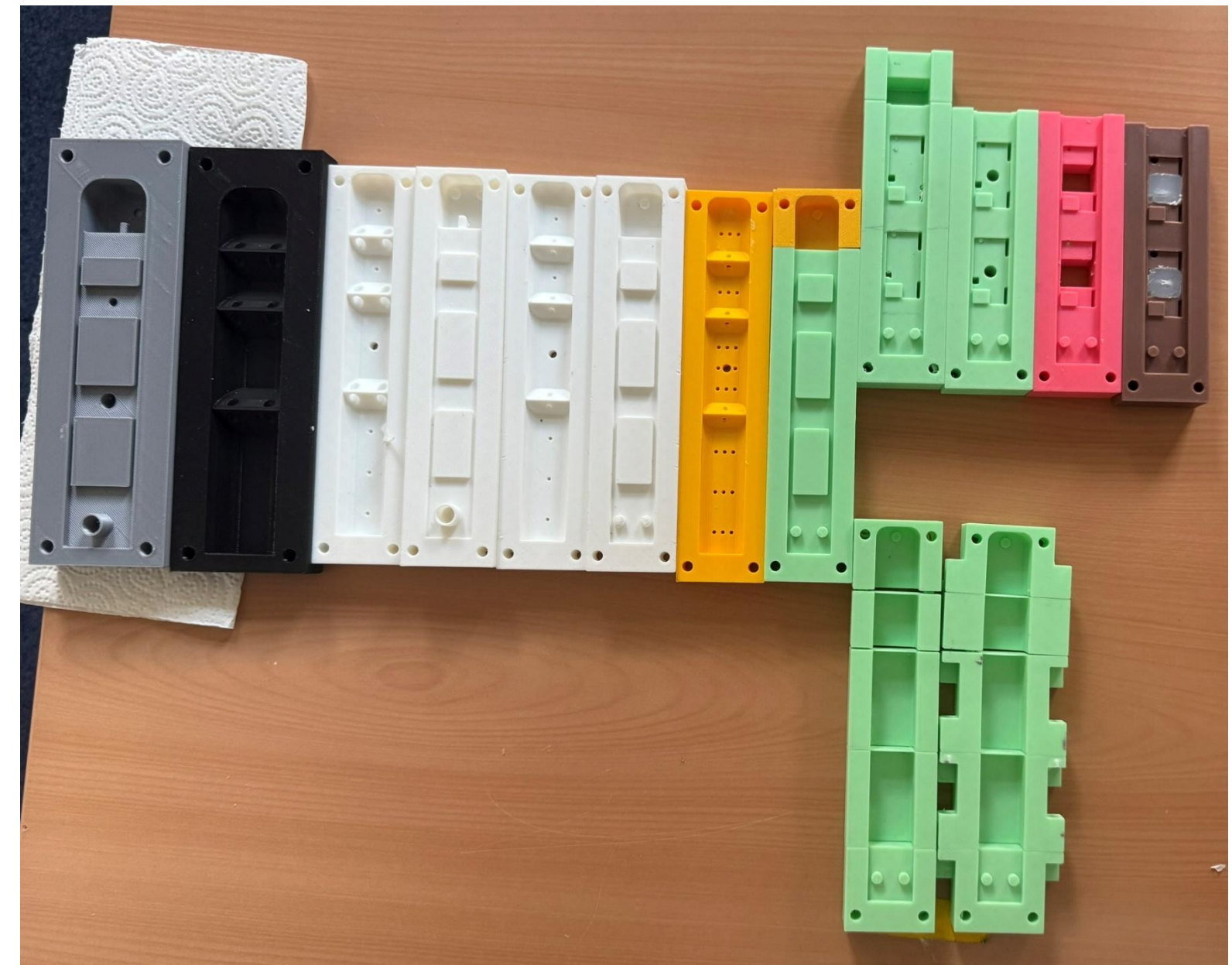


Algorithm 1 Generalized Joint Locking Control

```

1: Initialize digital output pins for Joint 1 and Joint 2
2: Start serial communication
3: while true do
4:   if a serial command is available then
5:     Read incoming command
6:     if command == '1' then
7:       Actuate(Joint 1, lock) {EM: set pin HIGH; LA: MoveSteps(N, forward)}
8:     else if command == '2' then
9:       Actuate(Joint 1, unlock) {EM: set pin LOW; LA: MoveSteps(N, backward)}
10:    end if
11:    if command == '3' then
12:      Actuate(Joint 2, lock) {EM: set pin HIGH; LA: MoveSteps(N, forward)}
13:    else if command == '4' then
14:      Actuate(Joint 2, unlock) {EM: set pin LOW; LA: MoveSteps(N, backward)}
15:    end if
16:    Print status message confirming action performed
17:  end if
18: end while
  
```

Versions





Driving Actuator vs the locking actuator

- Smaller
- less power consuming
- Needs less torque

Anatomy

- 1 : 1.5 : 2.6: 3.7 for the segments DP:MP:PP:M.
- Tip = $3.95 \pm 0.61\text{mm}$, DP= $17.40 \pm 1.85\text{mm}$, MP= $26.33 \pm 3.00\text{mm}$, PP= $44.63 \pm 3.81\text{mm}$, and M= $64.60 \pm 5.38\text{mm}$
- Finger length – 166mm

